

FACULTY OF ENGINEERING AND TECHNOLOGY
BACHELOR OF TECHNOLOGY

ICT WORKSHOP
(03010702ES01)

2nd SEMESTER
ELECTRONICS & COMMUNICATION ENGINEERING
DEPARTMENT

ICT WORKSHOP

PREFACE

It gives us immense pleasure to present the first edition of Workshop Practical Book for the B. Tech. 1st year students for **PARUL UNIVERSITY**.

The Workshop laboratory course at **PARUL UNIVERSITY, WAGHODIA, and VADODARA** is designed in such a way that students develop the basic understanding of the subject in the practical session and then try their hands on the experiments to realize and check the Workshop concepts learnt during the practical sessions. The main objective of the Electronics Workshop course is: To Understand every aspect of Electronics & Electrical components and its applications.

The objective of this ICT Workshop Laboratory Practical Book is to provide a comprehensive source for all the experiments included in the laboratory course.

We acknowledge the authors and publishers of all the books which we have consulted while developing this Practical book. Hopefully this *Workshop Practical Book* will serve the purpose for which it has been developed.

INSTRUCTIONS TO STUDENTS

1. Be prompt in arriving to the laboratory and always come well prepared for the experiment.
2. Be careful while working on the equipment's operated with high voltage power supply.
3. Work quietly and carefully. Give equal opportunity to all your fellow students to work on the instruments.
4. Every student should have his/her individual copy of the Workshop Practical Book.
5. Every student has to prepare the notebooks specifically reserved for the Workshop Practical work "Workshop".
6. Every student has to necessarily bring his/her Workshop Practical Book, Workshop Practical Class Notebook and Workshop Practical Final Notebook, when he/she comes to the practical to perform the experiment.
7. Record your observations honestly. Never makeup reading or doctor them to get a better fit on the graph or to produce the correct result. Display all your observations on the graph (If applicable).
8. All the observations have to be neatly recorded in the Workshop Practical Class Notebook (as explained in the Workshop Practical Book) and verified by the instructor before leaving the laboratory.
9. If some of the readings appear to be wrong then repeat the set of observations carefully.
10. Do not share your readings with your fellow student. Every student has to produce his/her own set of readings by performing the experiment separately.
11. After verification of the recorded observations, do the calculation in the Workshop Practical Class Notebook (as explained in the Workshop Practical Book) and produce the desired results and get them verified by the instructor.
12. Never forget to mention the units of the observed quantities in the observation table. After calculations, represent the results with appropriate units.
13. Calculate the percentage error in the results obtained by you if the standard results are available and also try to point out the sources of errors in the experiment.
14. Find the answers of all the questions mentioned at the end of each experiment in the Workshop Practical Book.
15. Finally record the verified observations along with the calculation and results in the Workshop Practical Note Book.
16. Do not forget to get the information of your next allotment (the experiment which is to be performed by you in the next laboratory session) before leaving the laboratory from the Technical Assistant.
17. The grades for the Workshop Practical course work will be awarded based on your performance in the laboratory, regularity, recording of experiments in the Workshop Practical Final Notebook, lab quiz, regular viva-voce and end-term examination.

CERTIFICATE

This is to certify that

Mr./Ms.....

With enrollment no.....has successfully completed his/her

Laboratory experiments in the ICT Workshop (03010702ES01) from
the department
of..... during the
academic year



Date of Submission:.....

Staff In charge:.....

Head Of Department:.....

INDEX

Sr. No	Experiment Title	Page No		Date of Performance	Date of Assessment	Marks (out of 10)	Sign
		From	To				
1	Introduction to Breadboard and Digital Multi Meter, Verify the circuit analysis (voltage and current)						
2	Identification, symbolic representation and testing of electronics basic components. (Diode, Zener diode, LED, Transistor) using Digital Multi Meter						
3	To study CRO and Function generator with specifications.						
4	To verify Ideal Value and practical value of voltage regulator ICs.						
5	Introduction to Arduino IDE with LED Blinking.						
6	Interfacing Button-Controlled LED with Arduino board. LED control using a push button.						
7	Buzzer Alarm System Using Arduino.						
8	Demonstrate the function of LCD Display Interfacing with Arduino.						
9	Relay-Based ON/OFF Control System with Arduino						
10	Interfacing Temperature Sensor (DHT11) with Arduino.						
11	Demonstrate the working of Ultrasonic Distance Measurement sensor.						
12	Verify the functionality of water flow sensor using Arduino.						
13	Demonstrate the working of PIR Sensor using Arduino.						
14	Implement a Mini Project using Electronics Components and Sensors.						

EXPERIMENT NO. 1

AIM: Introduction to Breadboard and Digital Multi Meter, Verify the circuit analysis (voltage and current)

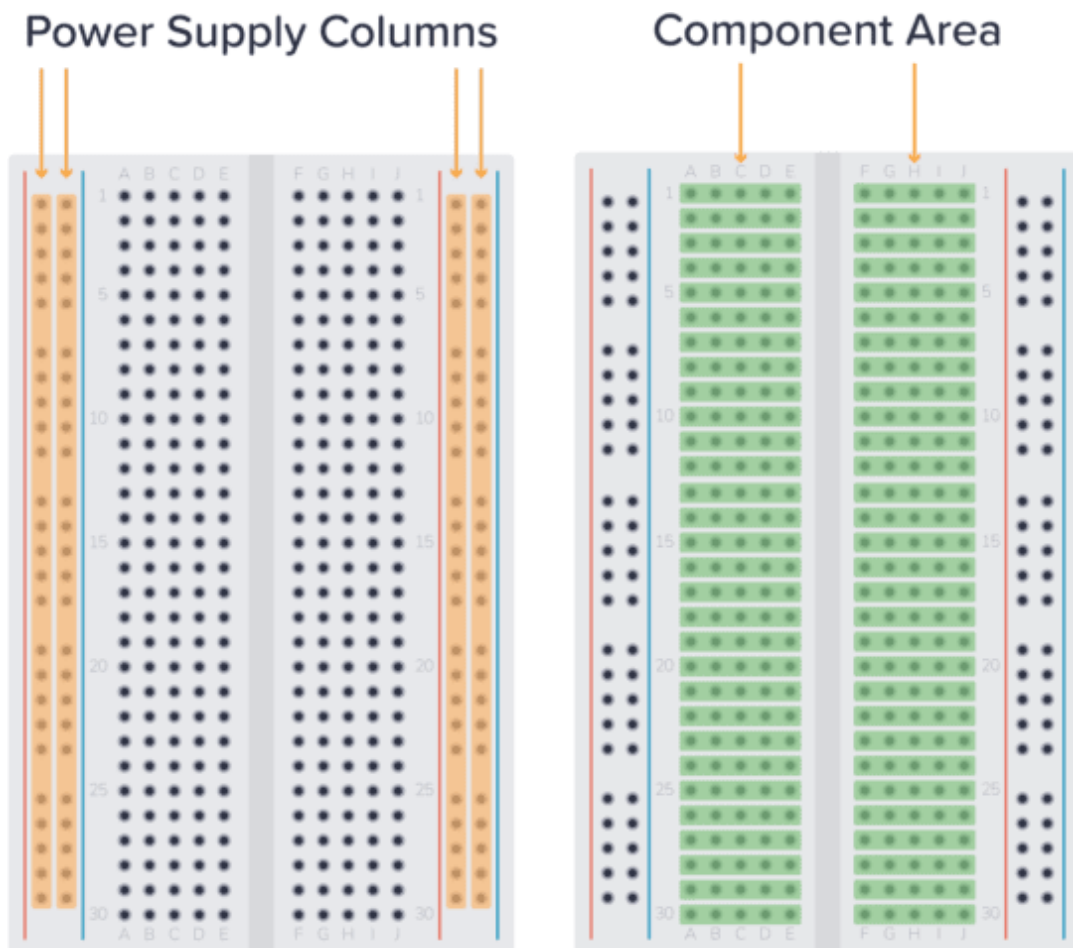
APPARATUS:

THEORY:

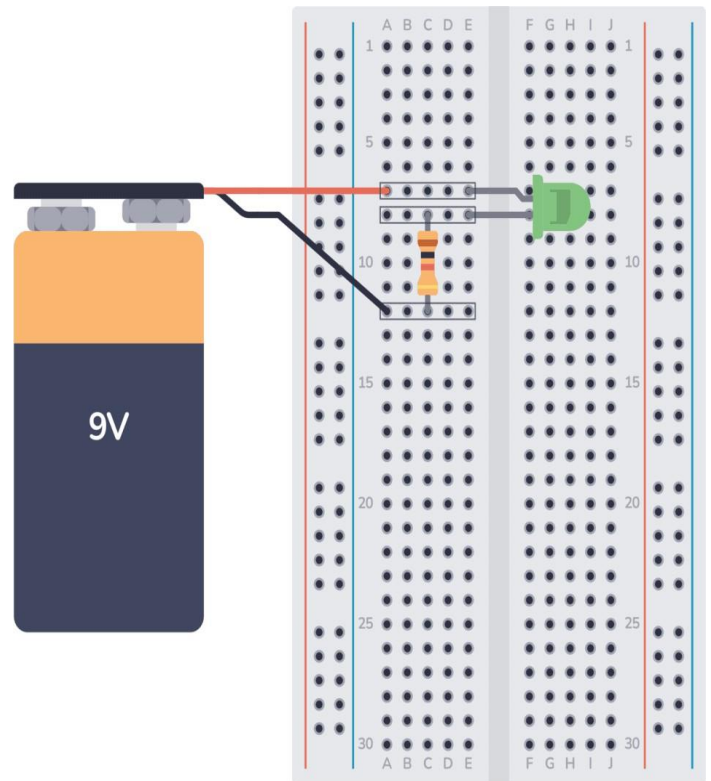
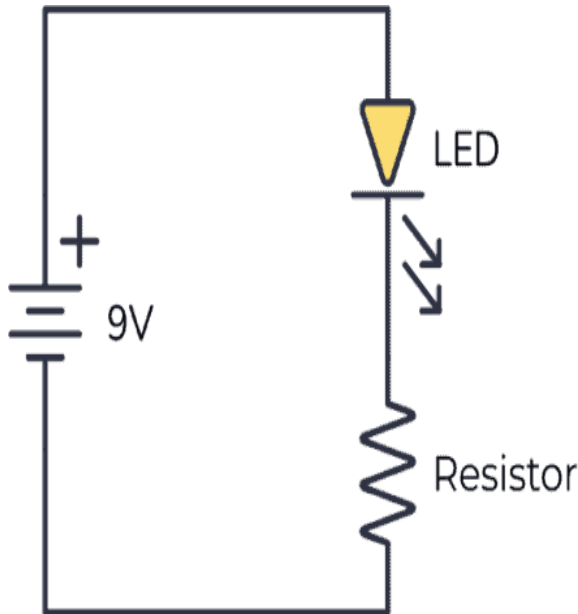
Introduction to Breadboard

A breadboard is a simple and useful tool for connecting a circuit. It's really useful for beginners as you can easily experiment and test out circuits without soldering. The breadboard has internal connections between its holes. Some vertical connections and some horizontal connections.

Normally, you use the columns on the sides to connect your power supply. And you use the rows in the middle to connect your components. In the middle of the board, you'll find the component area. This is where you normally connect your components. Here, the holes are connected horizontally, in rows. And the left side is separated from the right side. This means that the pins of your component must be on different rows, or on different sides of the board



Example Breadboard Circuit



Introduction To Multimeter:

A Multimeter is an instrument which measures electrical parameters such as AC or DC voltage, current, and resistance. A Multimeter combines a voltmeter, an ammeter, and an ohmmeter. The two main kinds of a Multimeter are analog and digital.

A digital Multimeter has an LCD screen that displays the value of the parameter being measured. In analog Multimeter, a needle moves through a graduated scale the pointer reading shows the value of the parameter measured.

Digital Multimeter

Digital Multimeter (DMM) has a digital or liquid crystal display (LCD), where measurement readings appears numerical values. The DMM has a rotatory knob or selector switch allows to select the unit and range of measurement. We must first set the dial to the appropriate measurement. There are three or four terminals available on the multi-meter, one is common, second is volt or resistance, third and fourth terminal is for current measurement.

The various electrical parameters can be measured by selecting the required function and selecting the proper range. Key process that occurs within a digital multimeter for any measurement that takes place is that of voltage measurement. All other measurements are derived from this basic measurement. One of the most widely used ADC in digital multimeter is known as the successive approximation register or SAR. Some SAR ADCs may only have resolution levels of 12 bits, but those used in test equipment including DMMs generally have 16 bits or possibly more dependent upon the application.

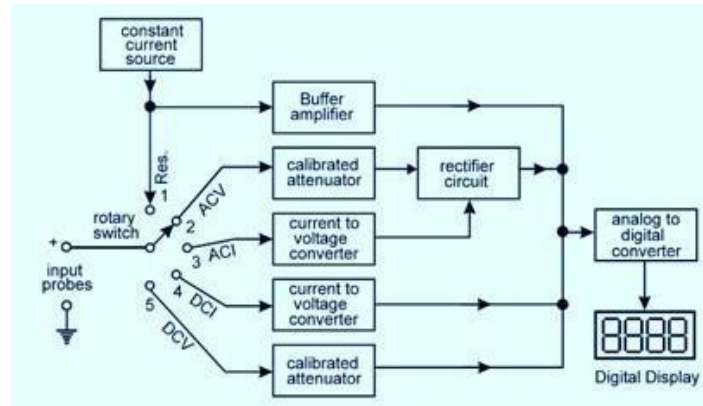


Fig 1. Block Diagram of Digital Multimeter

For the measurement of resistance Connect an unknown resistor across its input probes. Keep rotary switch in the position-1. The proportional current flows through the resistor, from constant current source. According to Ohm's law voltage is produced across it. This voltage is directly proportional to its resistance. This voltage is buffered and fed to A-D converter, to get digital display in Ohms

To measure AC voltage Connect an unknown AC voltage across the input probes. Keep rotary switch in position-2. The voltage is attenuated, if it is above the selected range and then rectified to convert it into proportional DC voltage. It is then fed to A-D converter to get the digital display in Volts. AC current is indirectly measured by converting it into proportional voltage. Connect an unknown AC current across input probes. Keep the switch in position-3. The current is converted into voltage proportionally with the help of I-V converter and then rectified. Now the voltage in terms of AC current is fed to A-D converter to get digital display in Amperes.

The DC current is also measured indirectly. Connect an unknown DC current across input probes. Keep the switch in position-4. The current is converted into voltage proportionally with the help of I-V converter. Now the voltage in terms of DC current is fed to A-D converter to get the digital display in Amperes.

To measure DC voltage Connect an unknown DC voltage across input probes. Keep the switch in position-5. The voltage is attenuated, if it is above the selected range and then directly fed to A-D converter to get the digital display in Volts.



Fig 2. Front Panel of Digital Multimeter

Front Panel Description of Digital Multimeter:

1. Probes: All the multimeter has two probes, one of Red colour and another of Black colour. The Black colour probe is used as common or negative probe while Red probe is used as Positive probe.
2. Common (Negative) Jack:- This jack has mark with term 'COM' and used to connect the Black probe as shown in diagram above. This probe is required for all measurements and test.
3. Positive Jack:- This jack is marked with sign of Voltage (V), Resistance (Ω) and Current (mA) and used to connect the Red probe.
4. Selector/ Rotatory Switch:- This switch or knob perform two important functions of Multimeter
 - (a) On/ Off switch of Multimeter.
 - (b) Selection of unit and range for the measurement and test.
5. LCD Display:-The display panel is used to display the measured values in numeric figures. The LCD display also displays some warning and alert symbols.
6. ACV: While moving clockwise (may be different with different models of Multimeter) the first unit is ACV; which means A.C. Voltage it is represented by wavy line.
7. DCA:-This part of digital Multimeter specified for measurement of D.C. Current; the range of current can be selected with rotatory knob. Next two blocks are used for special test related to amplification and frequency.
8. Continuity Test: the continuity test block is used for testing of some electrical and electronics devices. It simply shows that the two terminals allow the flow of electric current or not.
9. Resistance Range: A sign of Omega is used to represent the block of multimeter used for measurement of resistance. In this block Omega sign is termed as Ohm and 'k' represents range of Kilo- ohm.
10. DCV:-This block of multimeter is used to measure the D.C. Voltage represented by a bar above the dotted line.

Measurement of AC Voltage

The ac voltage is first converted into dc using rectifier. This dc voltage is converted into small signal using voltage limiter. This small dc voltage signal is given to the Analog to Digital converter. It converts the analog signal into the digital signal. Then it will be displayed on the display.

Measurement of DC Voltage

The dc voltage is given to the voltage limiter to convert into equivalent small signal. Then it is given to A to D converter. It converts the analog signal into Digital signal. Then it will be displayed on the display.

Measurement of Current

The unknown current is passed through I to V converter (i.e. current to voltage converter) in which it will flow through the low shunt resistance. It produces the voltage drop which is proportional to the current. This voltage drop is converted into digital signal using A to D converter. It converts the analog signal into digital signal. The reading will be displayed on the display.

Measurement of Resistance

To measure the resistance, the known current is passed through the unknown resistance using constant current source. The voltage drop across the resistance is applied to the A to D converter. It converts the analog signal into digital signal. Then it will be displayed on the display.

General Specifications of DMM:

Ranges:

For digital multimeter ranges varies according to the manufacturing company and also according to the parameter measured.

In multimeter ranges for AC / DC voltage, AC /DC current (mA & A) ranges, resistance ranges are mentioned. Auto and manual option is also present to read out parameters.

Accuracy:

In digital multimeter accuracy is usually specified in terms of percentage of reading plus a percentage of full-scale value, sometimes expressed in counts rather than percentage terms. Standard portable digital multimeters are specified to have an accuracy of typically 0.5% on the DC voltage ranges. Resolution:

The resolution of a multimeter is the smallest part of the scale which can be shown, which is scale dependent. For example, a multimeter that has a 1 mV resolution on a 10 V scale can show changes in measurements in 1mV increments.

The resolution of a multimeter is often specified in the number of decimal digits resolved and displayed. If the most significant digit cannot take all values from 0 to 9 is often termed a fractional digit. For example, a multimeter which can read up to 19999 (plus an embedded decimal point) is said to read 4½ digits.

CIRCUIT DIAGRAM

OBSERVATION TABLE:

- Measurement of Resistor**

Sr. No.	Theoretical Value	Practical Value

- Measurement of Voltage:**

	Theoretical Value	Practical Value
V _a		
V _b		
V _c		

- Measurement of Current:**

	Theoretical Value	Practical Value
I _a		
I _b		
I _c		

- Testing of Transistor:**

Transistor IC number	PNP/NPN

- Continuity Check:**

EXPERIMENT NO. 2

AIM: Identification, symbolic representation and testing of various electronics components.

APPARATUS:

THEORY:

Some of Electronic components used in electronic circuits are: - Diodes, Transistors, LED'S Photo diodes, IC'S, Rectifying bridges.

Diodes

Diodes are usually made from semiconductor materials, Silicon and Germanium being the most common. In electronics, a diode is a two-terminal electronic component that conducts electric current in only one direction. The term usually refers to a semiconductor diode, the most common type today. Diodes are fabricated from semi-conducting material. They have two leads: cathode (k) and an anode (A). The most important property of all diodes is their resistance is very low in one direction and very large in the opposite direction.

When a diode is placed in a circuit and the voltage on the anode is higher than the cathode, it acts like a low value resistor and current will flow. If it is connected in the opposite direction it acts like a large value resistor and current does not flow. In the first case the diode is said to be "forward biased" and in the second case it is "reverse biased." The symbolic representation of diode is shown in below figure 1.

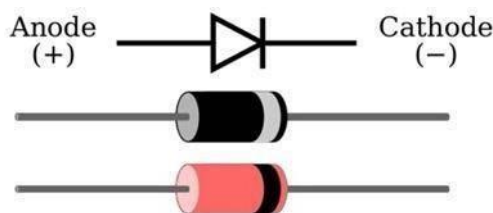


Fig 1. Symbolic representation of diode

The unidirectional behavior of diode is called rectification, and is used to convert alternating current to direct current. However, diodes can have more complicated behavior than this simple on- off action. These are exploited in special purpose diodes that perform many different functions. For example, specialized diodes are used to regulate voltage (Zener diodes), to electronically tune radio and TV receivers (varactor diodes), to generate radio frequency oscillations (tunnel diodes), and to produce light (light emitting diodes).

Testing of Diode using Multimeter:

Diodes can be tested using a multi meter. It is normally the resistance of the diode in both forward and reverse directions that is tested. There are however a number of points to remember when testing diodes. Digital Multimeters can test diodes using one of two methods:

Identification, symbolic representation and testing of various electronics components.
(Diode, Zener diode, LED, Transistor)

1. Diode Test mode: almost always the best approach.
2. Resistance mode: typically used only if a Multimeter is not equipped with a Diode Test mode. Note: In some cases it may be necessary to remove one end of the diode from the circuit in order to test the diode. Things to know about the Resistance mode when testing diodes:
 - Does not always indicate whether a diode is good or bad.
 - Should not be taken when a diode is connected in a circuit since it can produce a false reading.
 - Can be used to verify a diode is bad in a specific application after a Diode Test indicates a diode is bad.

A diode is best tested by measuring the voltage drop across the diode when it is forward-biased. A forward-biased diode acts as a closed switch, permitting current to flow.

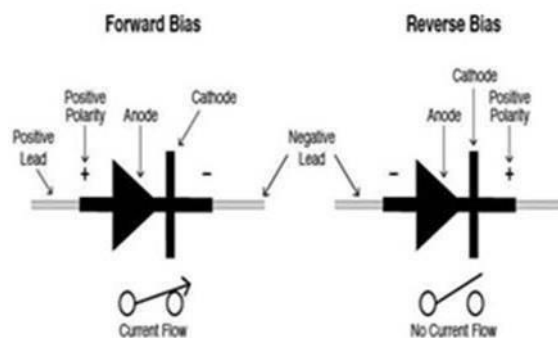


Fig 2. Diode Forward & Reverse Bias

A diode is forward-biased when the positive (red) test lead is on the anode and the negative (black) test lead is on the cathode. A Multimeter Diode Test mode produces a small voltage between test leads. The Multimeter then displays the voltage drop when the test leads are connected across a diode when forward-biased.

The Diode Test procedure is conducted as follows:

1. Make certain a) all power to the circuit is OFF and b) no voltage exists at the diode. Voltage may be present in the circuit due to charged capacitors. If so, the capacitors need to be discharged. Set the multimeter to measure ac or dc voltage as required.
2. Turn the dial (rotary switch) to Diode Test mode (). It may share a space on the dial with another function.
3. Connect the test leads to the diode. Record the measurement displayed.
4. Reverse the test leads. Record the measurement displayed.

Diode test analysis:

- A good forward-biased diode displays a voltage drop ranging from 0.5 to 0.8 volts for the most commonly used silicon diodes. Some germanium diodes have a voltage drop ranging from 0.2 to 0.3 V.
- The multimeter displays OL when a good diode is reverse-biased. The OL reading indicates the diode is functioning as an open switch.
- A bad (opened) diode does not allow current to flow in either direction. A multimeter will display OL in both directions when the diode is opened.
- A shorted diode has the same voltage drop reading (approximately 0.4 V) in both directions.

- A multimeter set to the Resistance mode (Ω) can be used as an additional diode test or, as mentioned previously, if a multimeter does not include the Diode Test mode.

The forward-biased resistance of a good diode should range from 1000 Ω to 10 M Ω . The resistance measurement is high when the diode is forward-biased because current from the multimeter flows through the diode, causing the high-resistance measurement required for testing. The reverse-biased resistance of a good diode displays 0L on a multimeter. The diode is bad if readings are the same in both directions.

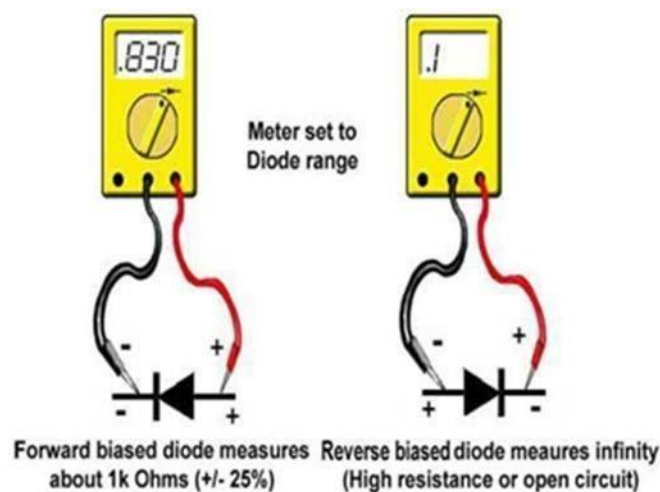


Fig 3. Diode Forward & Reverse Bias measurement using Multimeter

The resistance mode procedure is conducted as follows:

1. Make certain a) all power to the circuit is OFF and b) no voltage exists at the diode. Voltage may be present in the circuit due to charged capacitors. If so, the capacitors need to be discharged. Set the multimeter to measure ac or dc voltage as required.
2. Turn the dial to Resistance mode (Ω). It may share a space on the dial with another function.
3. Connect the test leads to the diode after it has been removed from the circuit. Record the measurement displayed.
4. Reverse the test leads. Record the measurement displayed.
5. For best results when using the Resistance mode to test diodes, compare the readings taken with a known good diode.

Light Emitting Diodes (LED's):

LEDs (Light Emitting Diodes) are constructed from a crystalline substance that emits light when a current flows through it. Depending on the crystalline material: red, yellow, green, blue or orange light is produced.

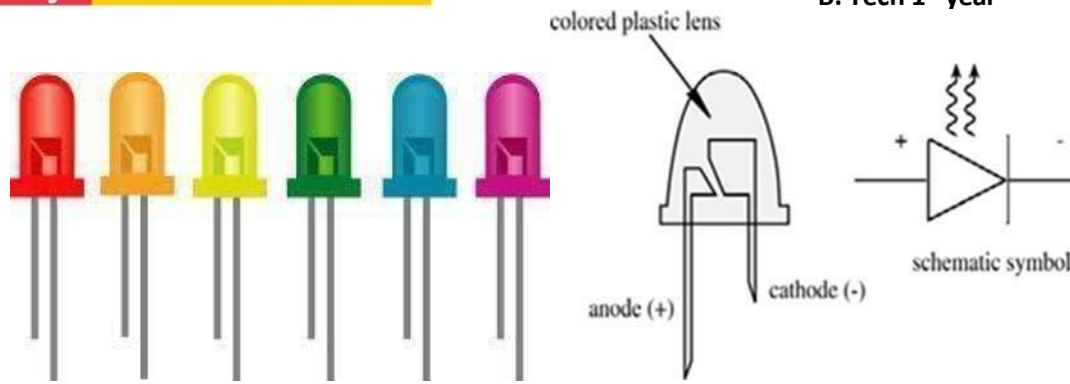


Fig 4. Types of LED & symbol

The cathode lead is identified on the body by a flat-spot on the side of the LED. The cathode lead is the shorter lead. LED can be tested in a similar way as that of p-n junction diode.

Zener Diode:

A Zener diode is a silicon diode that is optimized to conduct when it is reverse-biased. This article presents ways to identify one.

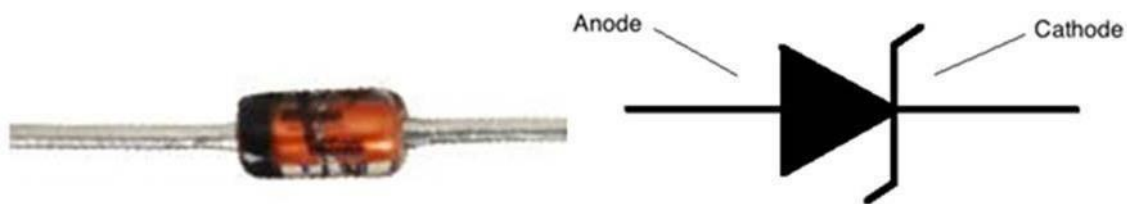


Fig 5. Zener diode and its symbol

A Zener diode is a silicon diode that is optimized to work in what is known as the breakdown region. This means that they are able to conduct when they are reverse-biased. This is unlike normal diodes, which will self-destruct. A Zener diode's breakdown voltages can range from 2 to 200 volts, making it useful in a variety of applications.

One popular usage is that of a voltage regulator. This is due to the Zener's ability to maintain constant output voltages when there are current changes in the circuit. This makes Zener's ideal as inputs to other circuits, or as voltage references for op-amps.

Testing of Zener diode using multimeter

1) Anode-Cathode Diode Resistance Test:

Using multimeter select the resistance function place the positive probe on the anode of the diode and the negative probe on the cathode of the diode (the black strip), as shown Figure 6 below. The diode hundred thousands of ohms should read a moderately low resistance, maybe a few Meter may read around 450kΩ.



Fig 6. Anode-Cathode diode resistance test



Fig 7. Cathode- Anode diode resistance test

2)Cathode-Anode Diode Resistance Test

Using the same test set up as mentioned above now reverse the connections such that the positive probe of the multimeter is now on the cathode of the diode and the negative lead on the anode refer figure 7. In this condition the diode should read a much higher resistance, well over $1\text{M}\Omega$. A typical reading may, for example, be $2.3\text{M}\Omega$. The multimeter may even indicate 'OL' for an open circuit, since the resistance is so high. A healthy Zener diode should read relatively low resistance in the forward biased direction and very high resistance in the reverse biased direction.

Testing of Zener diode using voltage function of multimeter

A second test to check a Zener diode is to measure its voltage with a voltmeter of a multimeter. Feed voltage to the Zener diode in reverse bias in series with a resistor. The voltage must be higher than the rated Zener voltage of the Zener diode.

As shown in below figure 8 Zener diode and a $1\text{k}\Omega$ resistor are supplied 9Volts from 9V battery.

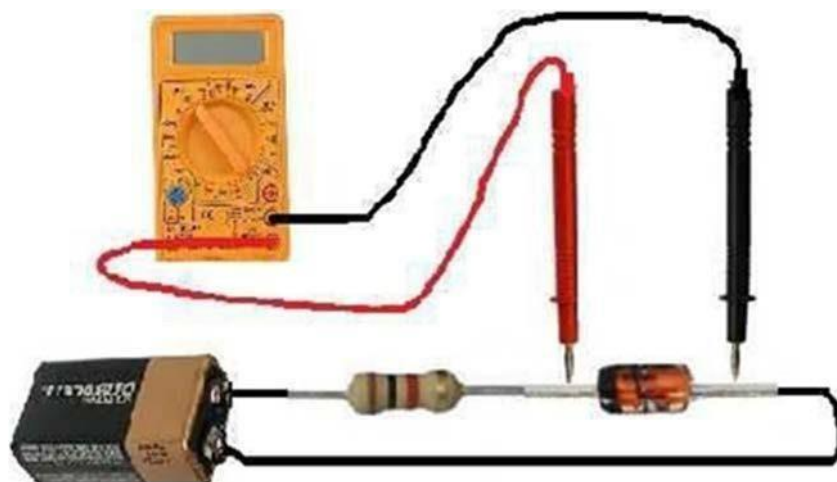


Fig 8. Voltage test of Zener diode

In above figure the rated voltage of Zener diode is 5.1V and the applied voltage is 9V. When measuring the voltage across the Zener diode, it must measure a voltage near its rated Zener voltage. If a Zener diode reads a voltage near its rated Zener voltage, V_Z , then the diode is healthy.

Transistors

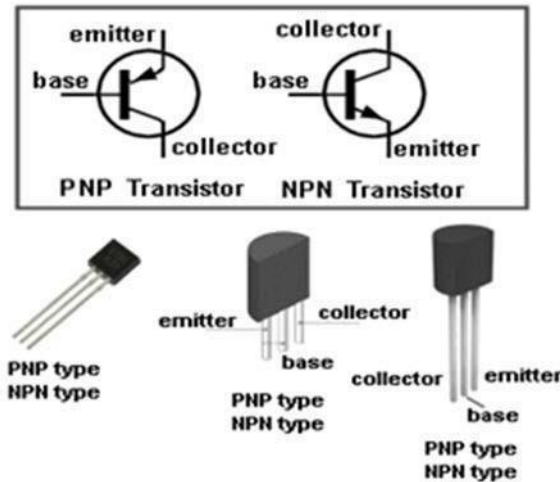


Fig 9. Bipolar Transistor symbol

Transistors are active components and are found every- where in electronic circuits. They are used as amplifiers and switching devices. As amplifiers, they are used in high and low frequency stages, oscillators, modulators, detectors and in any circuit needing to perform a function. In digital circuits they are used as switches. The most common type of transistor is called bipolar and these are divided into NPN and PNP types. Their construction-material is most commonly silicon or germanium Bipolar transistors have three leads: for base (B), emitter (E), and for collector (C).

How to test a BJT:

A digital multimeter is used to test the base to emitter and base to collector PN junction of the BJT. By using this test, you can also identify the polarity of an unknown device. PNP and NPN transistor can be checked using the digital multimeter.

The digital multimeter consists of two leads: black and red. Connect the red (positive) lead to the base terminal of the PNP transistor, and the black (negative) lead to the emitter or base terminal of the transistor. The voltage of a healthy transistor should be 0.7V, and the measurement across the emitter collector should read 0.0V. If the measured voltage is around 1.8V, then the transistor will be dead. Similarly, connect the black lead (negative) to the base terminal of the NPN transistor, and red lead (positive) to the emitter or collector terminal of the transistor. The voltage of a healthy transistor should be 0.7V, and the measurement across the emitter collector should read 0.0V. If the measured voltage is around 1.8V, then the transistor will be dead.

Step 1: Transistor Function on Multimeter

If the model number is not known, it is impossible to de- tail about the type of transistor just by looking at it. To ease the process, multimeters come with a function where one can calculate HFE or small signal Forward Current Gain of a bipolar transistor.



Fig 10. Transistor function on Multimeter

Apart, from detailing with the forward gain value, the HFE function solves two problems.

- Type of the transistor can be found.
- Collector, Base, Emitter terminals can also be known.

Step 2: Checking Transistor

1) Testing an NPN Transistor.

Place the three legs of the Transistor first in the E-B-C formation. The multimeter will show the reading if this formation is correct, else it will show over range. If the previous method doesn't work, place the three legs of the

Transistor in the B-C-E formation. If multimeter shows reading then it's correct. The reading is nothing but the HFE of that transistor.



Fig 11. Testing of NPN Transistor

2) Testing PNP Transistor.

Place the three legs of the transistor in the same manner as discussed above. After placing the transistor in the correct formation, multimeter would show the HFE value of the transistor.

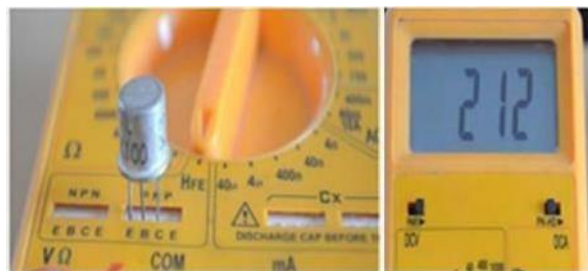


Fig 12. Testing of PNP Transistor

Uni -Junction Transistor:

The Unijunction Transistor or UJT for short, is another solid state three terminal device that can be used in gate pulse, timing circuits and trigger generator applications to switch and control either thyristors and triacs for AC power control type applications. Like diodes, unijunction transistors are constructed from separate P-type and N-type semiconductor materials forming a single (hence its name UniJunction) PN-junction within the main conducting N-type channel of the device.

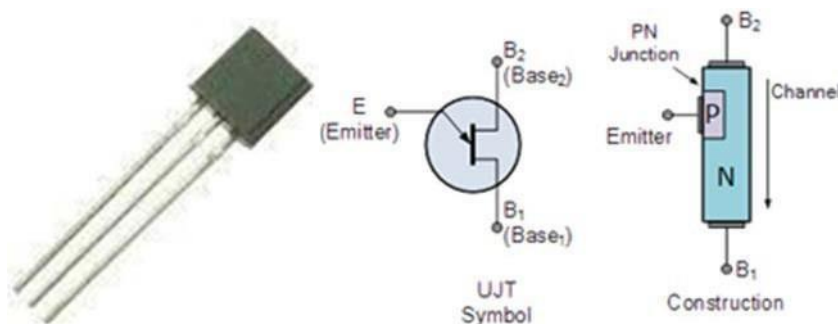


Fig 13. Unijunction transistor symbol and construction

Although the Unijunction Transistor has the name of a transistor, its switching characteristics are very different from those of a conventional bipolar or field effect transistor as it cannot be used to amplify a signal but instead is used as a ON-OFF switching transistor. UJT's have unidirectional conductivity and negative impedance characteristics acting more like a variable voltage divider during breakdown.

Like N-channel FET's, the UJT consists of a single solid piece of N-type semiconductor material forming the main current carrying channel with its two outer connections marked as Base 2 (B2) and Base 1 (B1). The third connection, confusingly marked as the Emitter (E) is located along the channel. The emitter terminal is represented by an arrow pointing from the P-type emitter to the N- type base.

How to test a UJT:

UJT (Uni junction transistor) can be easily tested by using a digital Multimeter. The three steps for testing the health of a UJT are as follows.

1. Measuring the resistance between B1 and B2 terminals. Set your digital Multimeter in resistance mode. Connect the positive lead of multimeter to the B1 terminal and negative lead to the B2 terminal. The Multimeter will show a high resistance (around 4 to 10K). Now connect the positive lead to B2 terminal and negative lead to B1 terminal. Again the Multimeter will show a high resistance (around 4 to 10K). Also both the readings will be almost same.

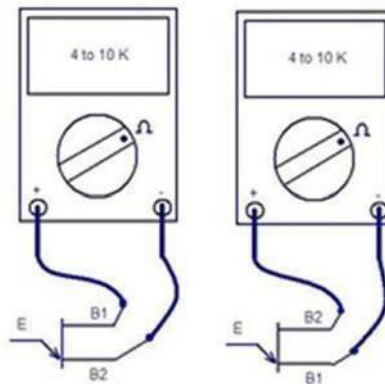


Fig 14. Measurement of Resistance

1. Reverse biasing the emitter junction.

Set the digital Multimeter in resistance mode. Connect negative lead of the multimeter to the emitter and positive lead to the B1. The multimeter will show a high resistance (around 100's of K's). Now connect the negative lead once again to the emitter and positive lead to B2. Again the meter will show a high resistance. In both cases the reading will be almost same.

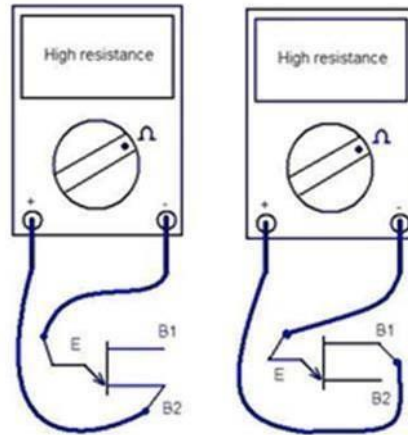


Fig 15. Reverse Biasing Emitter Junction

2. Forward biasing the emitter junction.

Set the digital Multimeter in resistance mode. Connect the positive lead to the emitter and negative lead to B1. The multimeter will show a low resistance (around few 100 ohms). Now connect the positive lead once again to the emitter and negative lead to the B2 terminal. Again the Multimeter will show a low resistance reading (around few 100 ohms). In both cases the reading will be almost same. This test is almost like forward biasing a diode.

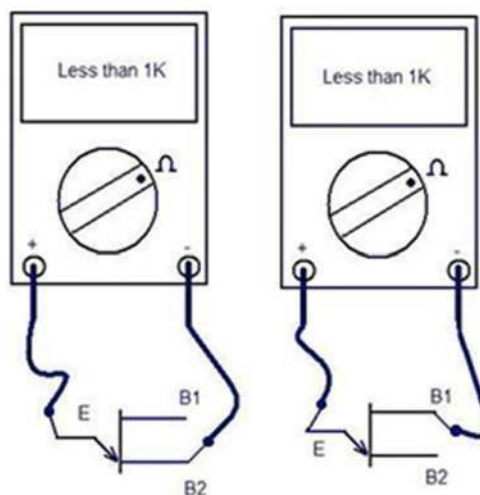


Fig 16. Forward Biasing Emitter Junction

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

EXPERIMENT NO. 3

AIM: To study CRO and Function generator with specifications.

APPARATUS:

THEORY:

Introduction to CRO

The cathode ray oscilloscope is a versatile laboratory instrument. If a laboratory has only CRO in it, other measuring instruments may not be required. This is the importance of CRO in scientific laboratories. With it we can measure, AC/DC voltage, AC/DC current, resistance, phase and phase difference between two or more waveforms, relative frequency of a waveform, observe the amount of noise present on a signal, etc.

In addition, CRO is also useful to observe the shape of waveform or signal and observe its real time progression on time axis. The waveform displayed on it, is observed with respect to x-y axes or co- ordinate system. The screen of CRO is plotted in terms of a measuring scale, known as graticule. Using this scale, the amplitude and wavelength of waveform can be accurately measured in centimeters and then converted into required unit.

Basic working principle of CRO

The cathode ray is a beam of electrons which are emitted by the heated cathode (negative electrode) and accelerated toward the fluorescent screen. The assembly of the cathode, intensity grid, focus grid, and accelerating anode (positive electrode) is called an electron gun. Its purpose is to generate the electron beam and control its intensity and focus. Between the electron gun and the fluorescent screen are two pair of metal plates - one oriented to provide horizontal deflection of the beam and one pair oriented to give vertical deflection to the beam. These plates are thus referred to as the horizontal and vertical deflection plates. The combination of these two deflections allows the beam to reach any portion of the fluorescent screen. Wherever the electron beam hits the screen, the phosphor is excited and light is emitted from that point. This conversion of electron energy into light allows us to write with points or lines of light on an otherwise darkened screen. Some important terms before going into the details of functional block diagram of CRO.

Y-input: It is the main input of CRO, to which the input signal is connected. The waveform of this input signal is displayed on the screen of CRT.

Vertical attenuator1: It consists of RC voltage divider, which is marked on the CRO front panel as Volt/div control knob. Thus the 'gain' of CRO can be controlled with Volt/div knob.

Vertical amplifier: It is a set of preamplifier and main vertical amplifier. The input attenuator sets up the gain of vertical amplifier.

Delay line: The delay line delays the striking of electron beam on the screen. It synchronizes the arrival of the beam on screen when time base generator signal starts sweeping the beam horizontally. The propagation delay² produced is about 0.25msec.

Trigger circuit: It takes the sample of input voltage connected at y-input of CRO and feeds it to the input of time base generator. So the TBG starts only when input signal is present at y-input.

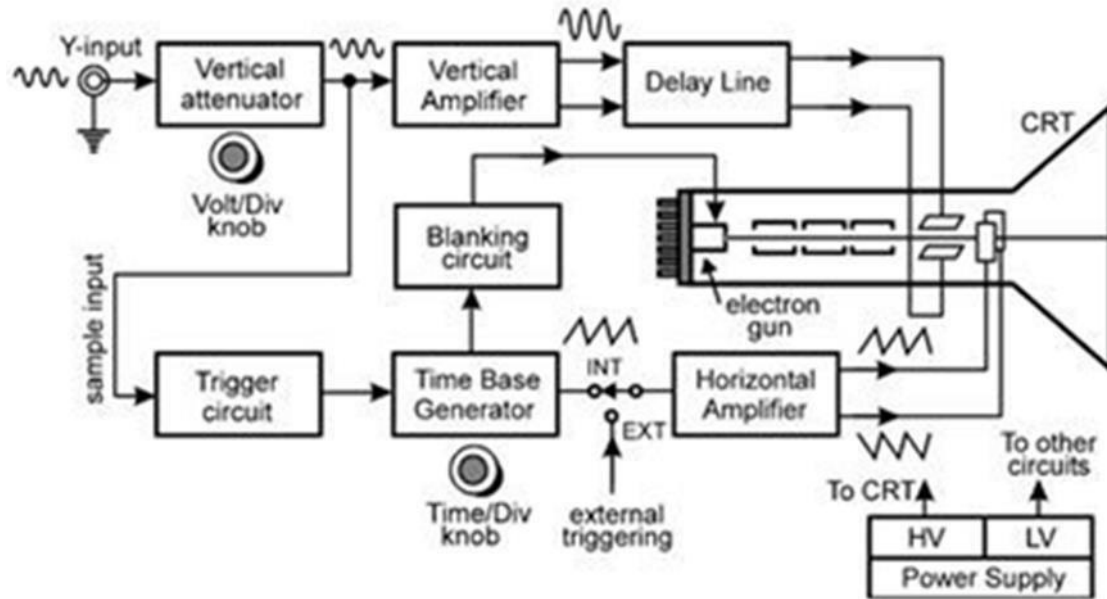


Fig 1. Block Diagram of Cathode Ray Oscilloscope

Time base generator: It produces a saw tooth wave. The waveform is used to sweep (move) the electron beam horizontally on the screen. The rate of rise of positive going edge of saw tooth wave- form is controlled by Time/div control knob. Thus, the saw tooth wave controls the horizontal deflection of electron beam along x-axis. A switch known as INT/EXT is also connected after the output of TBG. When the switch is in INT position, the output of TBG is connected to H-plates through horizontal amplifier. When it is in EXT position, internal saw tooth is cut-off and some external signal can be connected to horizontal plates.

Horizontal amplifier: It amplifies the saw tooth waveform coming from TBG. It contains phase in[-verter circuit also. Due to this circuit, two outputs are produced. One output produces positive going saw tooth and other output produces negative going saw tooth. The first output is connected to right side H-plate and the second output is connected to left side H-plate. So the electron beam moves properly from left to right of the screen.

Blanking circuit: It is necessary to eliminate the retrace, which would produce when the spot on screen moves from right to left. This retrace can produce confusion with the original wave. So when the electron beam reaches right end of screen, the negative blanking voltage is produced by TBG. It is fed to control grid of CRT, to stop the electron beam completely.

HV/LV power supply: The high voltage section is used to power the electrodes of CRT and the low voltage section is used to power the electronic circuits of the CRO.

Front Panel Controls:

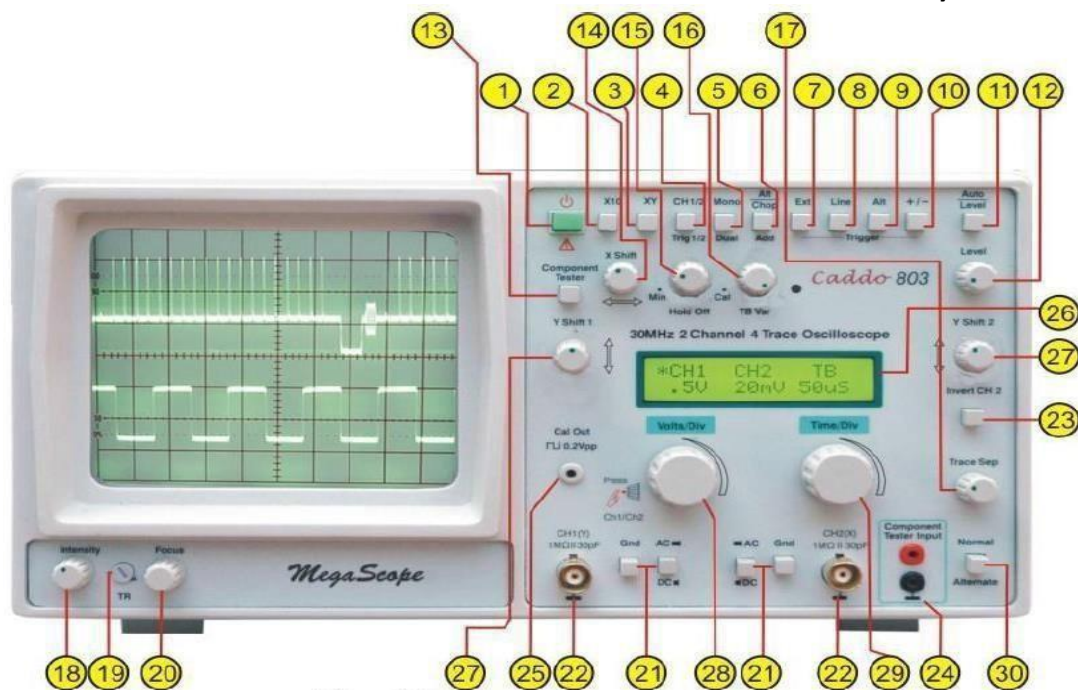


Figure 2 Front Panel of CRO

Sr. No.	Control	Function
1	Power 'On/Off':	Push buttons switch for supplying power to instrument.
2	x10	Switch when pushed gives 10 times magnification of the X signal.
3	XY	Switch when pressed cuts off the time base & allows access to the external horizontal signal to be fed through CH2 (used for X-Y display).
4	CH 1/2	Switch selects channel & trigger source (released Trig 1/2 CH1 & pressed CH2).
5	Mono/ Dual	Switch selects Mono or Dual trace operation.
6	Alt/ Chop/ Add	Switch selects alternate or chopped in Dual mode. If Mono is selected then this switch enables addition or subtraction of channel i.e. CH1 $\pm$ CH2.
7	Ext	Switch when pressed allows external triggering signal to be fed from the socket marked Trigger Input (20).
8	Line	Switch when pressed displayed signal synchronized with mains line frequency.
9	Alt	Selects alternate trigger mode from CH1 & CH2. In this mode both

		the signals are synchronized.
10	Slope (+/-)	Switch selects the slope of triggering, whether positive going or negative going.
11	Auto/Level	Switch selects Auto/Level position. Auto is used to get trace when no signal is fed at the input. In Level position the trigger level can be varied from the positive peak to negative peak with Level Control.
12	Level	Controls the trigger level from peak to peak amplitude of signal.
13	Component	Switch when pressed starts CT operation.
14	X Shift	Controls horizontal position of the trace.
15	TB Var	Controls the time speed in between two steps of Time/Div switch. For calibration put this pot fully anticlockwise at Cal position. Lamp glows for uncalibrated position.
16	Trace Sep	Trace Separator x1 & x10 in 4 trace operation (Alt).
17	Intensity	Controls the brightness of the trace.
18	TR	Trace Rotation controls the alignment of the trace with graticule (screw driver adjustment).
19	Focus	Controls the sharpness of the trace.
20	DC/AC/GND	Input coupling switch for each channel. In AC the signal is coupled through 0.1MFD capacitor.
21	CH 1 (Y) &	BNC connectors serve as input connection for CH 2 (X): CH1 & CH2. Channel 2 input connector also serves as horizontal external Input.
22	Invert CH2	Switch when pressed invert polarity of CH2.
23	Component Tester Input	To test any components in the CT mode, put one test prod in this socket and connect the other test prod in ground socket
24	Cal Out	Socket provided for square wave output 200mV used for probe
25	Hold off	Controls Hold Off time between sweeps. Used for Stable Triggering of composite signals.
26	Digital Readout	LCD window for displaying Digital Readout for Volt/Div. & Time/Div. settings.
27	Y Pos 1 & 2	Controls provided for vertical deflection of trace for each channel.
28	Volts/Div CH1	Switch selects Volt/Div. step for CH1 & CH2

29	Time /Div	Switch selects Time/Div. steps.
30	Normal/Alternate :	Switch selects Normal (x1) or Alternately expanded (x1 & x10) simultaneous positions.

Product Features of a typical analog CRO are as under:

- Dual channel
- Bandwidth up-to 30MHz.
- Invert facility in both channels.
- Vertical deflection coefficients: 5mV to 20 V / div.
- Time base: 20ns-0.2 s / div.
- Variable Hold –Off.
- X10 magnification.
- Triggering capability.
- Saw tooth output (5Vpp approx.)

Measurements using CRO

The primary uses of the cathode ray oscilloscope (CRO) are to measure voltage, to measure frequency and to measure phase.

Measuring voltage

Because of its effectively infinite resistance, the CRO makes an excellent voltmeter. It has a relatively low sensitivity, but this can be improved by the use of an internal voltage amplifier. The oscilloscope must first be calibrated by connecting a dc. source of known

e.m.f. to the Y-plates and measuring the deflection of the spot on the screen. This should be repeated for a range of values, so that the linearity of the deflection may be checked. The known e.m.f. is then connected and its value found from the deflection produced. Most oscilloscopes have a previously calibrated screen giving the deflection sensitivity in volts per cm or volts per scale division. In this case a calibration by a d.c. source may be considered unnecessary.

Measuring frequency

Using the calibrated time base the input signal of unknown frequency may be 'frozen', and its frequency found directly by comparison with the scale divisions. Alternatively the internal time base may be switched off and a signal of known frequency applied to the X-input. If the signal of unknown frequency is applied to the Y-input, Lissajous figures are formed on the screen. Analysis of the peaks on the two axes enables the unknown frequency to be found.

Measuring phase

The internal time base is switched off as above and two signals are applied as before. The frequency of the known signal is adjusted until it is the same as that of the unknown signal. An ellipse will then be formed on the screen and the angle of the ellipse will denote the phase difference between the two signals.

Function generator

Signal generators, also known variously as function generators, RF and microwave signal generators, pitch generators, arbitrary waveform generators, digital pattern generators or frequency generators are electronic devices that generate repeating or non-repeating electronic signals (in either the analog or digital domains). They are generally used in designing, testing, troubleshooting, and repairing electronic or electro acoustic devices.



Fig 3. Function Generator

A function generator is a device which produces simple repetitive waveforms. Such devices contain an electronic oscillator, a circuit that is capable of creating a repetitive waveform refer figure 4. The most common waveform is a sine wave, but saw tooth, step (pulse), square, and triangular waveform oscillators are commonly available as are arbitrary waveform generators (AWGs). If the oscillator operates above the audio frequency range (>20 kHz), the generator will often include some sort of modulation function such as amplitude modulation (AM), frequency modulation (FM), or phase modulation (PM) as well as a second oscillator that provides an audio frequency modulation waveform.

Arbitrary waveform generators

Arbitrary waveform generators, or AWGs, are sophisticated signal generators which allow the user to generate arbitrary waveforms, within published limits of frequency range, accuracy, and output level. Unlike function generators, which are limited to a simple set of waveforms; an AWG allows the user to specify a source waveform in a variety of different ways. AWGs are generally more expensive than function generators, and are often more highly limited in available band- width; as a result, they are generally limited to higher-end design and test applications. Some of the signal generators are RF, Analog, AF, Logic, Pitch and audio signal generators.

Working of Function Generator

Simple function generators usually generate triangular waveform whose frequency can be controlled smoothly as well as in steps. This triangular wave is used as the basis for all of its other outputs. The triangular wave is

generated by repeatedly charging and discharging a capacitor from a constant current source. This produces a linearly ascending or descending voltage ramp. As the output voltage reaches upper and lower limits, the charging and discharging is reversed using a comparator, producing the linear triangle wave. By varying the current and the size of the capacitor, different frequencies may be obtained. Saw tooth waves can be produced by charging the capacitor slowly, using a current, but using a diode over the current source to discharge quickly - the polarity of the diode changes the polarity of the resulting saw tooth, i.e. slow rise and fast fall, or fast rise and slow fall.

A 50% duty cycle square wave is easily obtained by noting whether the capacitor is being charged or discharged, which is reflected in the current switching comparator output. Other duty cycles (theoretically from 0% to 100%) can be obtained by using a comparator and the saw tooth or triangle signal. Most function generators also contain a non-linear diode shaping circuit that can convert the triangle wave into a reasonably accurate sine wave by rounding off the corners of the triangle wave in a process similar to clipping in audio systems.

A typical function generator can provide frequencies up to 20 MHz RF generators for higher frequencies are not function generators in the strict sense since they typically produce pure or modulated sine signals only.

Function generators, like most signal generators, may also contain an attenuator, various means of modulating the output waveform, and often the ability to automatically and repetitively "sweep" the frequency of the output waveform (by means of a voltage- controlled oscillator) between two operator-determined limits. This capability makes it very easy to evaluate the frequency response of a given electronic circuit.

Specifications

Typical specifications for a general-purpose function generator are:

- Produces sine, square, triangular, saw-tooth (ramp), and pulse output. Arbitrary waveform generators can produce waves of any shape.
- It can generate a wide range of frequencies.
- Good frequency stability analogue generators & digital generators.
- Maximum sine wave distortion of about 1% for analogue generators.
- AM or FM modulation may be supported.
- Output amplitude up to 10 V peak-to-peaks.
- Amplitude can be modified, usually by a calibrated attenuator with decade steps and continuous adjustment within each decade.
- Some generators provide a DC offset voltage, e.g. adjustable between -5V to +5V.
- An output impedance of 50 Ω .

PROCEDURE:

OBSERVATION TABLE:

Input Signal	Input value (from Function generator)			Measured value (from CRO)		
	Amplitude	Time	Frequency	Amplitude	Time	Frequency
Sine wave						
Square wave						
Triangular wave						

To Observe & Draw waveform of Lissajous Pattern:

Draw the Output pattern using Component Tester in CRO:

Short Circuit	
Open Circuit	
Resistor	

Capacitor

Diode

Zener Diode

CALCULATION:

EXPERIMENT NO. 4

AIM: To verify Ideal Value and practical value of voltage regulator ICs.

APPARATUS:

THEORY:

Introduction

A power supply is an electronic device that supplies electric energy to an electrical load. The primary function of a power supply is to convert one form of electrical energy to another and, as a result, power supplies are sometimes referred to as electric power converters.

Some types of power supplies are:-

- 1) D.C. power supply
- 2) A.C. power supply.
- 3) Switched-mode power supply
- 4) Uninterruptible power supply.

D.C power supply

- 1) Linear regulator

The function of a linear voltage regulator is to convert a varying DC voltage to a constant, often specific, lower DC voltage. In addition, they often provide a current limiting function to protect the power supply and load from over current (excessive, potentially destructive current).

A constant output voltage is required in many power supply applications, but the voltage will vary with changes in load impedance and will also vary with changing input voltage. To overcome this, some power supplies use a linear voltage regulator to maintain the output voltage at a steady value, independent of fluctuations in input voltage and load impedance. Linear regulators can also reduce the magnitude of ripple and noise present appearing on the output voltage.

- 2) Regulated A.C.to D.C. power supply

A regulated power supply is an embedded circuit; it converts unregulated AC into a constant DC. With the help of a rectifier it converts AC supply into DC. Its function is to supply a stable voltage (or less often current), to a circuit or device that must be operated within certain power supply limits. The output from the regulated power supply may be alternating or unidirectional, but is nearly always DC.

Construction and working of regulated D.C power supply

The basic components of dc power supply are transformer, rectifier, filter and voltage regulator. Starting with an ac voltage, a steady dc voltage is obtained by rectifying the ac voltage,

then filtering to a dc level and, finally, regulating to obtain a desired fixed dc voltage. The regulation is usually obtained from an IC voltage regulator unit, which takes a dc voltage and provides a somewhat lower dc voltage, which remain s the same even if the input dc voltage varies or the output load connected to the dc voltage changes.

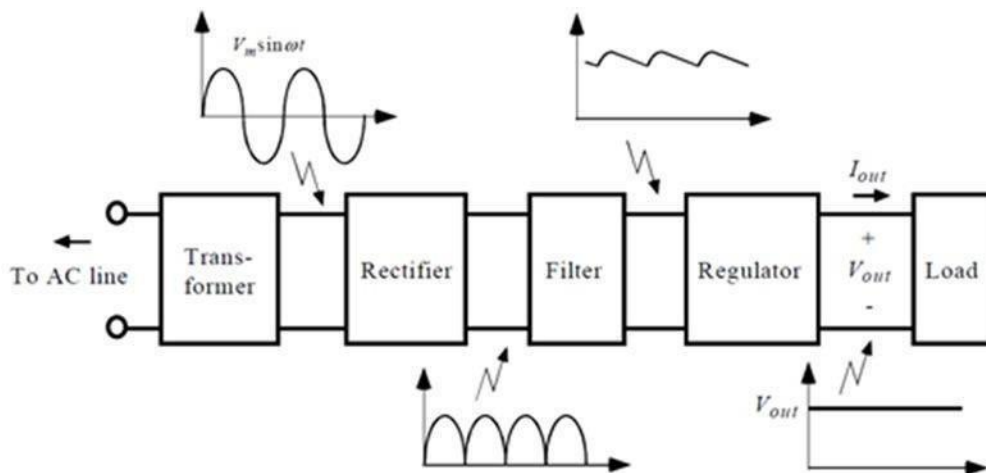


Fig 1. Components of a typical linear power supply

A block diagram containing the parts of a typical power supply and the voltage at various points in the unit is shown in above figure 1. The ac voltage, typically 120 Vrms, is connected to a transformer, which steps that ac voltage down to the level for the desired dc output. A diode rectifier then provides a full-wave rectified voltage that is initially filtered by a simple capacitor filter to produce a dc voltage. This resulting dc voltage usually has some ripple or ac voltage variation. A regulator circuit can use this dc input to provide a dc voltage that not only has much less ripple voltage but also remains the same dc value even if the input dc voltage varies somewhat or the load connected to the output dc voltage changes. This voltage regulation is usually obtained using one of a number of popular voltage regulator IC units.

Ripple factor: $(V_{ac} \text{ (rms)} / V_{dc}) \times 100\%$ Voltage Regulation

Another factor of importance in a power supply is the amount the dc output voltage changes over a range of circuit operation. The voltage provided at the output under no- load condition (no current drawn from the supply) is reduced when load current is drawn from the supply (under load). The amount the dc voltage changes between the no -load and load conditions is described by a factor called voltage regulation.

$$\text{Voltage Regulation} = \frac{\text{No- load voltage} - \text{Full load voltage}}{\text{Full load voltage}} \quad \dots \text{Eq 1}$$

$$\%VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\%$$

PROCEDURE:

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

SIGNATURE OF FACULTY WITH DATE

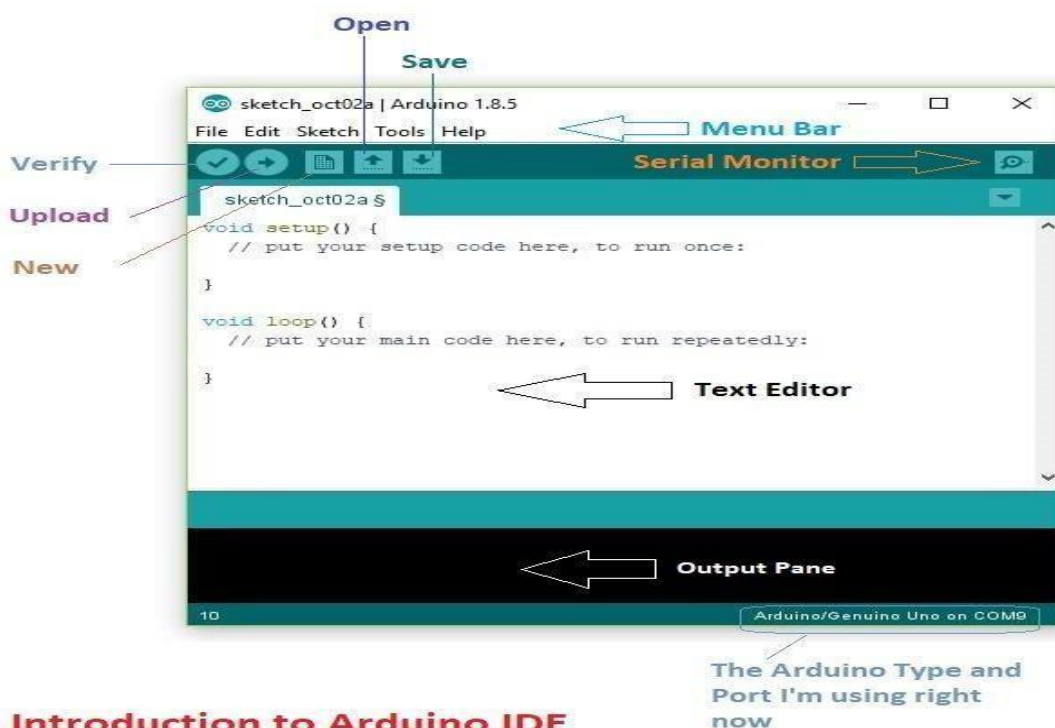
EXPERIMENT NO. 5

AIM: Introduction to Arduino IDE with LED Blinking

APPARATUS:

THEORY: The Arduino IDE (Integrated Development Environment) is a user-friendly platform for writing, compiling, and uploading code to Arduino boards, making it accessible for beginners and experienced developers alike.

- Arduino IDE is an open-source software, designed by Arduino.cc and mainly used for writing, compiling & uploading code to almost all Arduino Modules.
- It is an official Arduino software, making code compilation too easy that even a common person with no prior technical knowledge can get their feet wet with the learning process.
- It is available for all operating systems i.e. MAC, Windows, Linux and runs on the Java Platform that comes with inbuilt functions and commands that play a vital role in debugging, editing and compiling the code.
- A range of Arduino modules available including Arduino Uno, Arduino Mega, Arduino Leonardo, Arduino Micro and many more.
- Each of them contains a microcontroller on the board that is actually programmed and accepts the information in the form of code.
- The main code, also known as a sketch, created on the IDE platform will ultimately generate a Hex File which is then transferred and uploaded in the controller on the board.
- The IDE environment mainly contains two basic parts: Editor and Compiler where former is used for writing the required code and later is used for compiling and uploading the code into the given Arduino Module.
- This environment supports both C and C++ languages.



LED blinking using Arduino IDE

About LED:

Pinout

LED includes two pins:

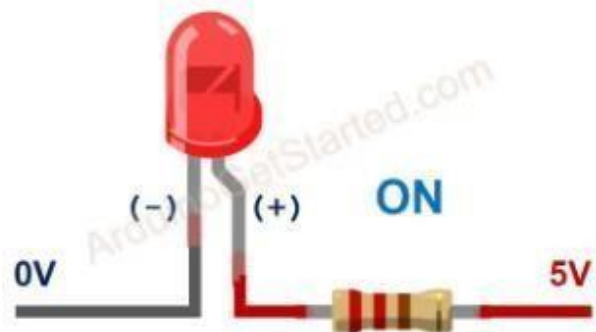
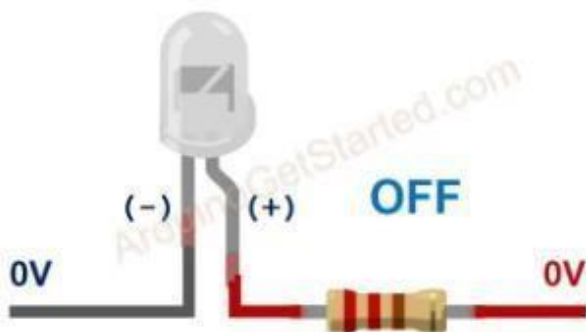
- ◆ Cathode(-) pin: needs to be connected to **GND**
- ◆ Anode(+) pin: is used to control LED's state



How It Works

After connecting the cathode(-) to **GND**:

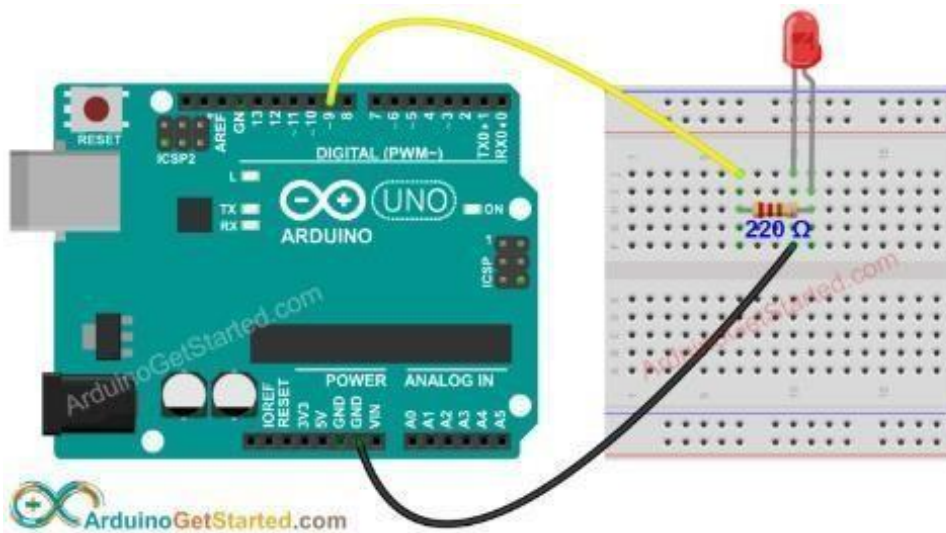
- ◆ If connecting **GND** to the anode(+), LED is **OFF**.
- ◆ If connecting **VCC** to the anode(+), LED is **ON**.



Arduino - LED

When an Arduino's pin is configured as a digital output, the pin's voltage can be programmatically set to **GND** or **VCC** value.

By connecting the Arduino's pin to LED's anode(+) pin (via a resistor), we can programmatically control LED's state.



PROCEDURE:

[illegible]

PERFORMANCE OUTPUT:

CONCLUSION:

Questions:

1. Which Arduino Board is used during experiment? Why?
2. Write the specifications of Arduino UNO board?
3. What are the basic components needed for LED interfacing with Arduino?
4. Write a code use for Blinking LED using Arduino.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

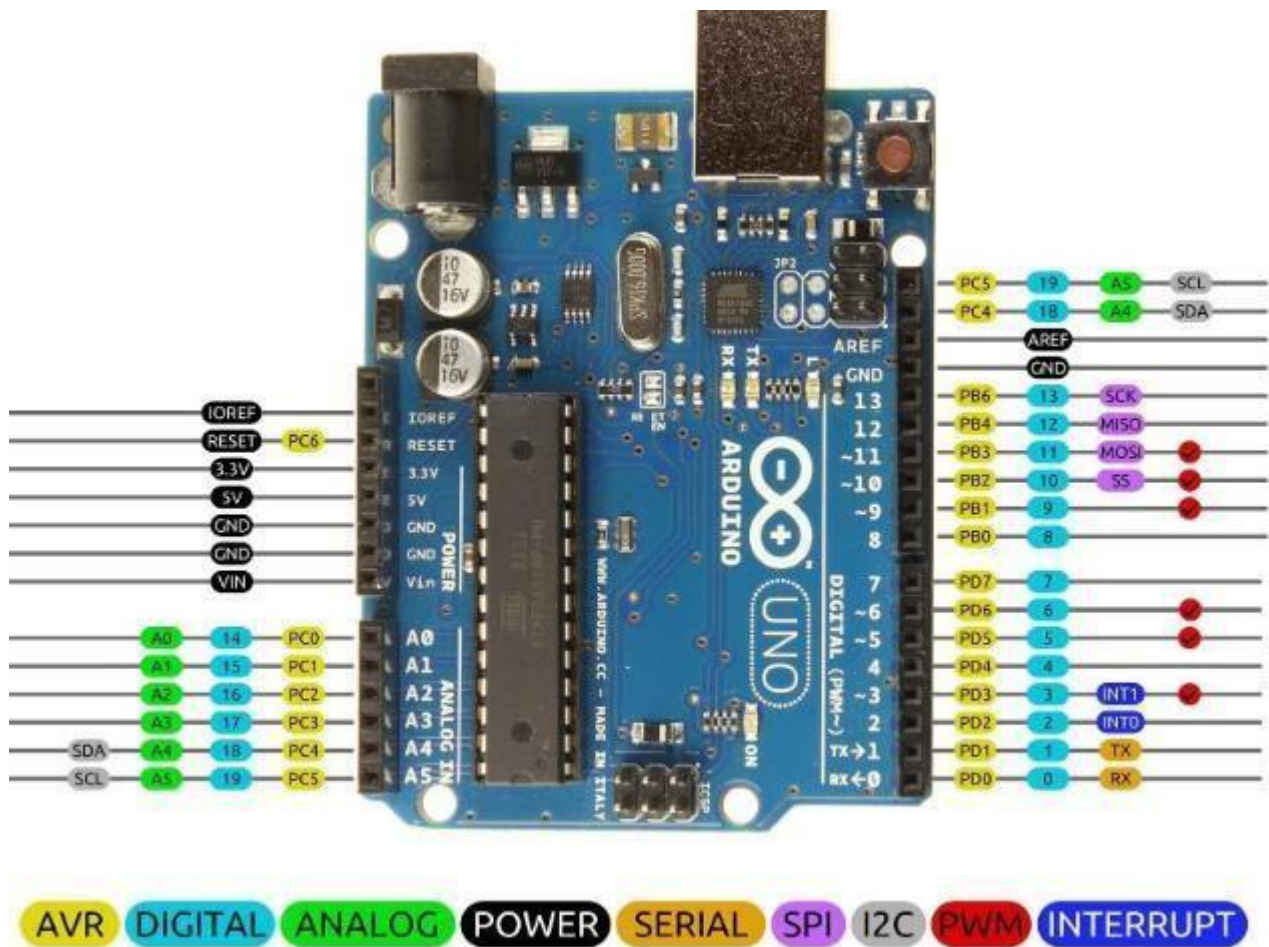
EXPERIMENT NO. 6

AIM: Interfacing Button-Controlled LED with Arduino board.

APPARTUS:

THEORY:

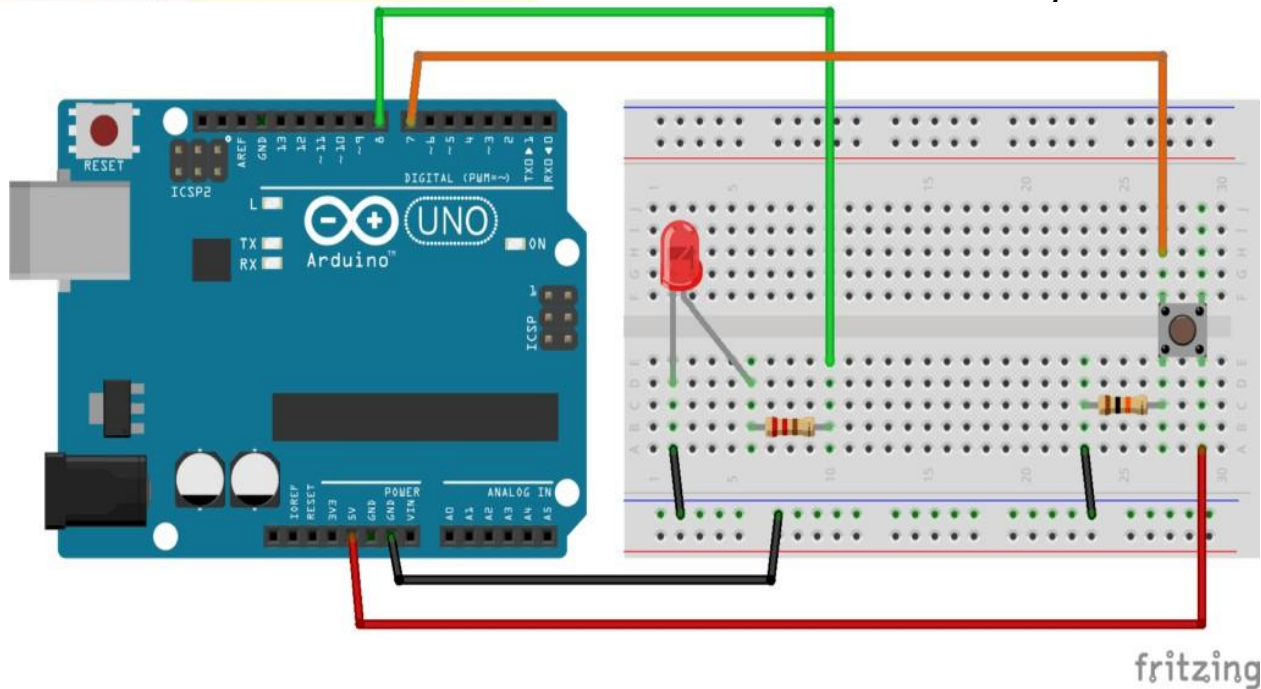
ARDUINO BOARD PIN LAYOUT



Arduino circuit with an LED and a button

To build the circuit you will need those components:

- Arduino board (any board, if you don't have Uno you can easily adapt by finding +
- Breadboard.
- LED – any color.
- Push button.
- 220 Ohm resistor for the LED. If you don't have this specific value, any resistor from 330 to 1k Ohm will do.
- 10k Ohm resistor for the push button. If you don't have, you can go until 20k-50k Ohm.
- A bunch of male to male wires (including if possible black, red, and other colors).



fritzing

Step by step instructions to build the circuit

- First, make sure to power off your Arduino – remove any USB cable.
- Plug a black wire between the blue line of the breadboard and a ground (GND) pin on the Arduino board.
- Plug the LED. You can notice that the LED has a leg shorter than the other. Plug this shorter leg to the ground (blue line here) of the circuit.
- Connect the longer leg of the LED to a digital pin (here pin no 8, you can change it). Add a 220 Ohm resistor in between to limit the current going through the LED.
- Add the push button to the breadboard, like in the picture.
- Connect one leg of the button to the ground, and put a 10k Ohm resistor in between. This resistor will act as a “pull down” resistor, which means that the default button’s state will be LOW.
- Add a red wire between another leg of the button and VCC (5V).
- Finally, connect a leg of the button (same side as the pull down resistor) to a digital pin (here 7).

PERFORMANCE RESULT:

CONCLUSION:

EXPERIMENT NO. 7**AIM: Buzzer Alarm System Using Arduino.****APPARTUS:****THEORY:****What is Buzzer Alarm System?**

An alarm system is a device that warns people about highly unsafe situations at the workplace or home. A buzzer alarm system is intended to warn us of an emergency by sending an auditory signal. Hence it uses a buzzer to create a sound.

About Buzzer

The buzzers are used to generate sound, beep or even melody of a song. They come in several types, categorized by distinct characteristics. Let's explore these categories based on the following characteristics:

- **Control Method:**
 - Active Buzzers
 - Passive Buzzers
- **Sound-Generating Mechanism:**
 - Piezo Buzzers
 - Regular Buzzers
- **Operating Voltage:**
 - Low Voltage (3-5V)
 - High Voltage (12V)

Piezo Buzzer:

- Uses piezoelectric crystals.
- Produces high-pitched, clear sounds.
- Versatile with various tones and frequencies.
- Efficient and commonly used in alarms and musical devices.

Regular Buzzer:

- Uses electromagnetic coils.
- Produces basic, buzzy sounds.
- Limited tone variation.
- Used in simpler applications like doorbells and basic alarms.

In the market, there is a 3V-24V active buzzer, which you can use as a 3-5V active buzzer or high voltage buzzer (12V...):

- When you connect this buzzer directly to an Arduino pin, it will create a normal sound → suitable for sound indicators like keypad sounds.

- When you connect this buzzer to a high voltage via a relay, it will create a loud sound → suitable for warning sounds.

Schematic

Pinout



3-5V Buzzer

- Connect directly to ESP32 => normal sound



3-24V Buzzer

- Connect directly to ESP32 => normal sound
- Connect to high-voltage via relay => loud sound



3-5V Buzzer Module

- Connect directly to ESP32 => normal sound

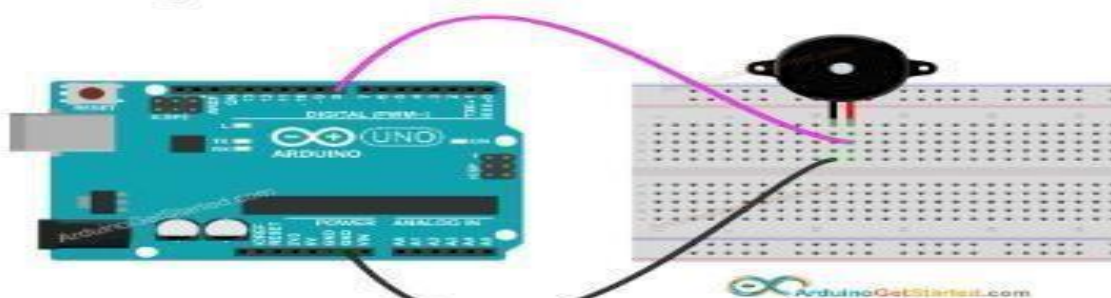
Buzzer Pinout

A standard buzzer typically has two pins:

- ◆ Negative (-) pin: Connect this pin to **GND** (0V).
- ◆ Positive (+) pin: This pin receives the control signal from the Arduino, either directly or via a relay.

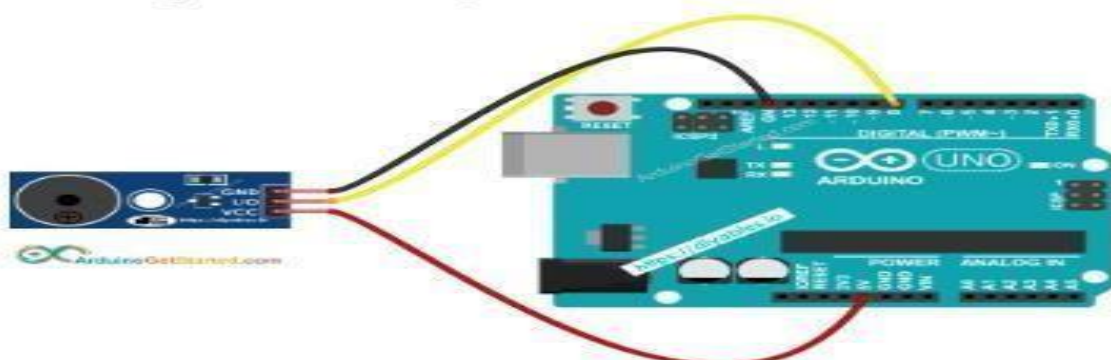
Wiring Diagram

◆ The wiring between Arduino and piezo buzzer



This image is created using Fritzing. Click to enlarge image.

◆ The wiring between Arduino and piezo buzzer module



PERFORMANCE RESULT:

1. What is the primary function of a buzzer in an alarm system?
2. Which type of buzzer is commonly used in electronic alarm systems?
3. What is the typical voltage range required to operate a piezoelectric buzzer?
4. What component is often used to control the activation of a buzzer in an alarm system?
5. Write a Program for sound a buzzer in delay.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

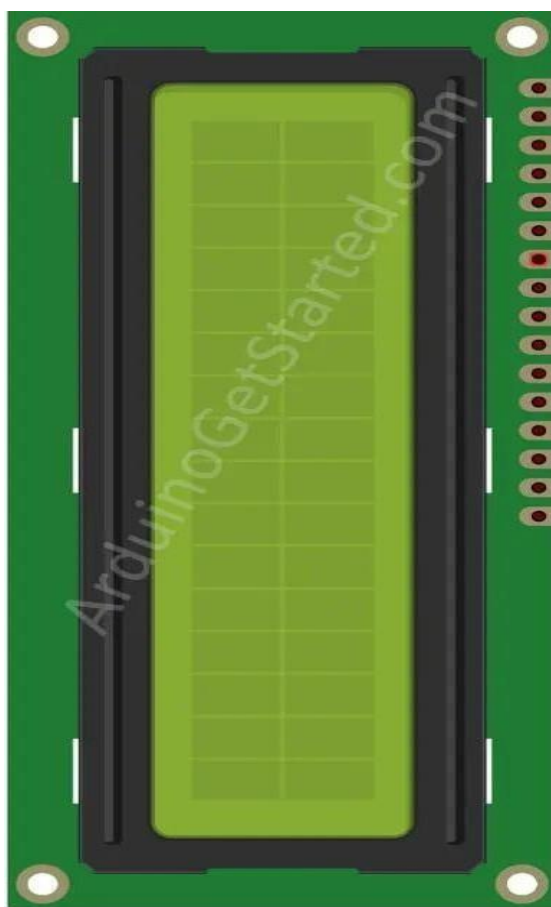
EXPERIMENT 8

AIM: Demonstrate the function of LCD Display Interfacing with Arduino..

APPARTUS:

THEORY:

About LCD 16x2



1	GND
2	VCC
3	Vo (Contrast)
4	RS (Register Select)
5	R/W (Read/Write)
6	EN (Enable)
7	D0
8	D1
9	D2
10	D3
11	D4
12	D5
13	D6
14	D7
15	A (Anode)
16	K (Cathode)

DATA pins



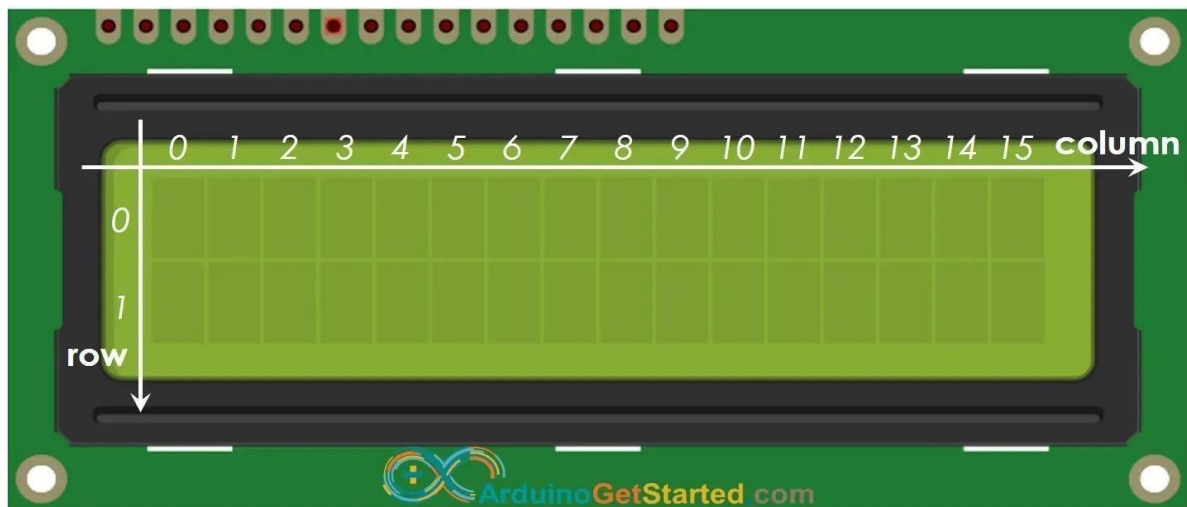
LCD has up to 16 pins. In the most common uses, we do NOT use all pins.

let's see these pins and their functionality:

- **GND pin:** needs to be connected to **GND** (0V).
- **VCC pin:** the power supply for the LCD, needs to be connected to **VCC** (5V).
- **Vo** (LCD Contrast) pin: controls the contrast and brightness of the LCD, can be connected to 5V (the highest contrast and brightness), or connected to a potentiometer (to adjust to the contrast and brightness).
- **RS** (Register Select) pin: There are two kinds of data that need to send to LCD: command (to control LCD) and data. These two are sent on the same data bus. RS pin tells the LCD whether the data on the data bus is the commands or the data.
 - If we want to send the command to control LCD, we need to set RS pin to **LOW** (like set the cursor to a specific location, clear the display ...).

- If we want to send the data to display on LCD, we need to set RS pin to **HIGH**.
- **R/W** (Read/Write) pin: is to select READ mode or WRITE mode.
 - If we want to read data from LCD, this pin needs to be set to **HIGH**.
 - If we want to send data to LCD, this pin needs to be set to **LOW**. Since we're just using this LCD as an OUTPUT device, we're going to tie this pin **LOW**.
- **EN** (Enable) pin: is used to enable the LCD. **HIGH** to enable the LCD, **LOW** to disable the LCD.
- **D0 – D7** (Data Bus) pins: carries data and command between Arduino and LCD. There are two modes to send data: 4-bit mode and 8-bit mode.
- **A – K** (Anode & Cathode) pins: are used to power the LCD backlight. A pin needs to be connected to **VCC**, K pin needs to be connected to **GND**.

This 16x2 includes 16 columns and 2Row



How It Works?

Sending Data to the LCD (for Display)

1. Arduino sets the RS pin to HIGH (selecting the data register).
2. Arduino writes data to the D4–D7 pins (data bus).
3. The LCD receives this data from the data bus.
4. Since RS is HIGH, the LCD stores the data in the **data register**, then displays it on the screen.

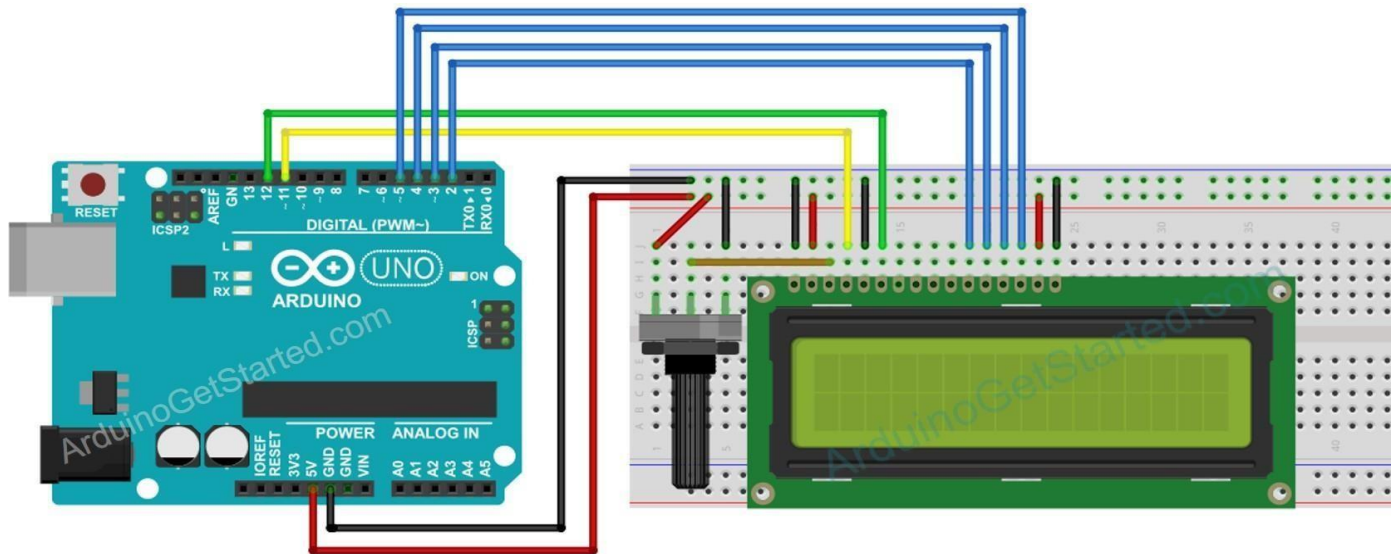
Sending Commands to the LCD (for Control)

Commands control functions like blinking, clearing the display, or setting the cursor position.

1. Arduino sets RS to LOW (selecting the command register).
2. Arduino writes the command to the D4–D7 pins (data bus).
3. The LCD receives this data from the data bus.
4. Since RS is LOW, the LCD stores the data in the **command register** and executes the command accordingly.

Arduino & LCD

Controlling an LCD can be complex, but the **Liquid Crystal** library simplifies the process. You don't need to handle low-level instructions—just connect your Arduino to the LCD and use the library's built-in functions!



PERFORMANCE RESULT:

EXPERIMENT NO. 9

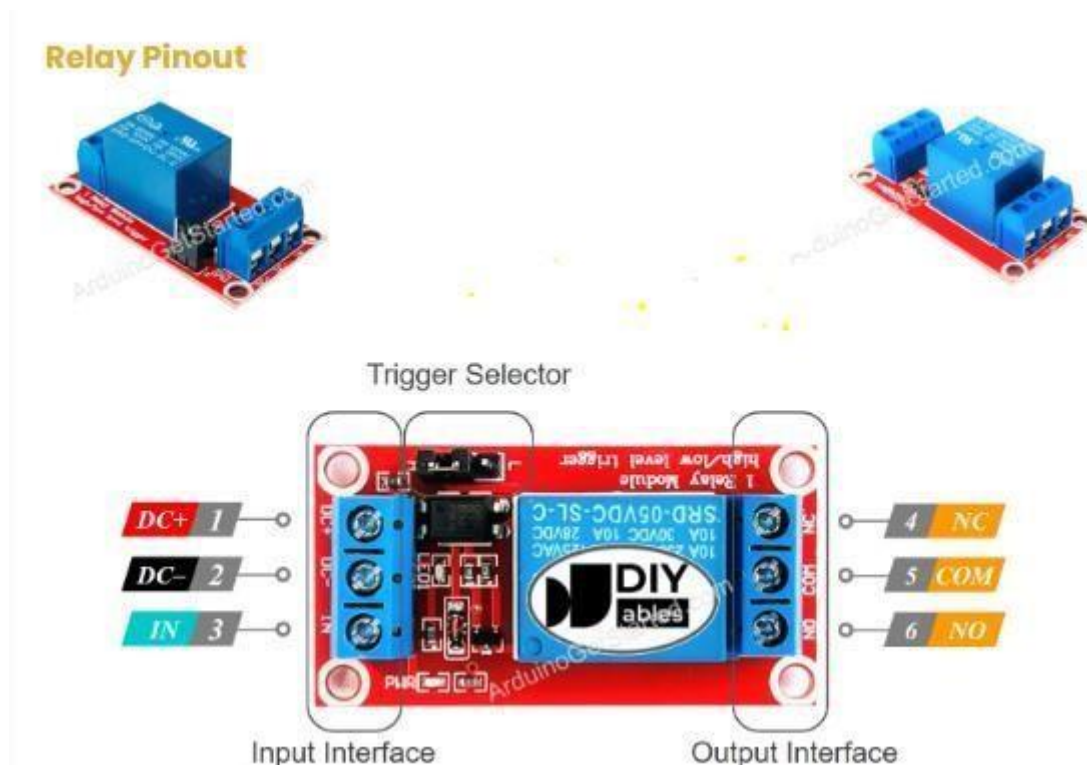
AIM: Relay-Based ON/OFF Control System with Arduino

APPARTUS:

THEORY:

About RELAY

A RELAY is a Programmable electrical Switch, Which can be controlled by Arduino or any Microcontroller. It is used to programmatically Control ON/OFF the devices. Which is used to high voltage or high current. It is Bridge between Arduino and High voltage Devices.



Relay has two groups of pins: input (low voltage) group and output (high voltage) group.

Pins in the input group are connected to Arduino, including three pins:

- **DC- pin:** needs to be connected to **GND** (0V)
- **DC+ pin:** needs to be connected to **VCC** (5V)
- **IN pin:** receives the control signal from Arduino

Pins in the output group are connected to the high voltage device, including three pins (usually in screw terminal):

- **COM pin:** is the common pin. It is used in both **normally open** mode and **normally closed** mode
- **NO pin:** is **normally open** pin. It is used in the **normally open** mode
- **NC pin:** is **normally closed** pin. It is used in the **normally closed** mode

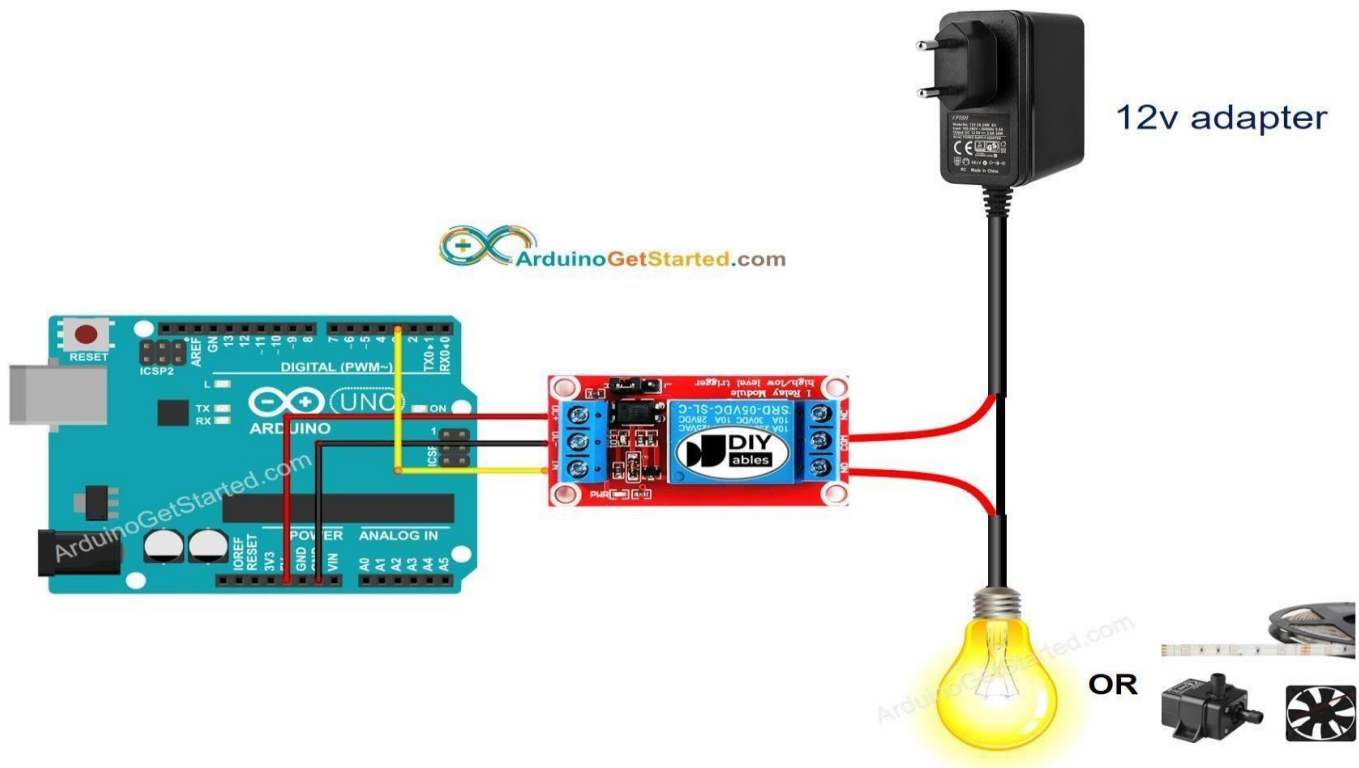
In practice, we usually do NOT use all of the pins in the high voltage group. We use only two of them:

We use only **COM pin** and **NO pin** if we use **normally open** mode.

We use only **COM pin** and **NC pin** if we use **normally closed** mode.

Additionally, if the relay supports both **LOW** and **HIGH** level triggers, there is usually a jumper to select one of two: **LOW** level trigger or **HIGH** level trigger.

Wiring Diagram



PERFORMANCE RESULT:

CONCLUSION: _____

QUESTIONS:

1. What the difference is between normally open (NO) and normally closed (NC) pins?
2. Which relay pin receives the control signal from the Arduino?
3. Write a step wise procedure to connect RELAY with Arduino
4. Write a Code for Switch ON/OFF by Relay controlled.
5. What is the difference between LOW level and HIGH level trigger relays?

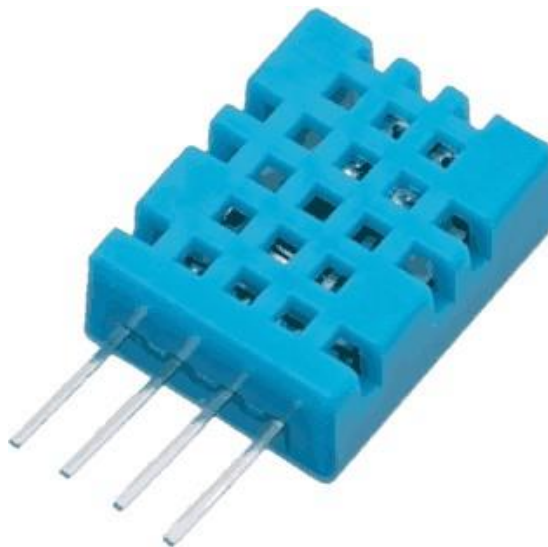
EXPERIMENT NO. 10**AIM: Demonstrate the working of Temperature Sensor-DHT11****APPARTUS:****THEORY:**

A Temperature Sensor is a piece of electronic equipment that detects the temperature of its surroundings and transforms the incoming data into electronic output data to control record or signal temperature variations. There are several types of temperature sensors. Some of them need direct contact with the physical target that is being identified, which is introduced as contact temperature sensors. In contrast, others sense the temperature of the targets indirectly, which is described as non-contact temperature sensors.

Temperature Sensor Working

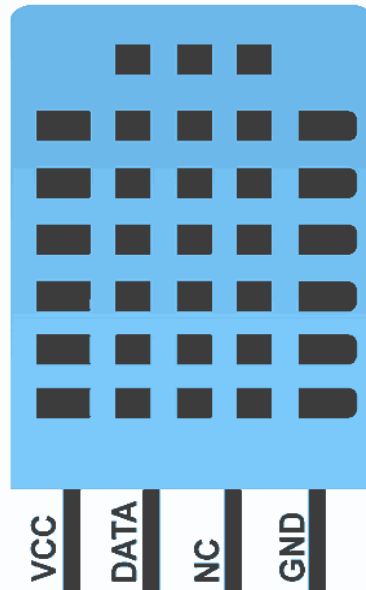
The fundamental working of this sensor is based on the voltage in its diode. The temperature variation is directly related to the resistance of this diode. The resistance of the diode is detected and transformed into simple and readable values of temperature such as Fahrenheit, Kelvin, or Centigrade and demonstrated in meaningful formats instead of readout values. These temperature sensors are employed to sense the internal temperature of various structures like power plants.

The fundamental principle of the temperature sensors operation is the voltage modification in the MOSFET terminals. If the voltage reduces, the temperature also decreases, according to the voltage drop between the emitter in the MOSFET and terminals of the base sensor.

DHT11

DHT11 humidity and temperature sensor

- The DHT11 is a basic, low cost digital temperature and humidity sensor.
- DHT11 is a single wire digital humidity and temperature sensor, which provides humidity and temperature values serially with one-wire protocol.
- DHT11 sensor provides relative humidity value in percentage (20 to 90% RH) and temperature values in degree Celsius (0 to 50 °C).
- DHT11 sensor uses resistive humidity measurement component, and NTC temperature measurement component.
- DHT11 is a 4-pin sensor, these pins are VCC, DATA, GND and one pin is not in use shown in fig below.



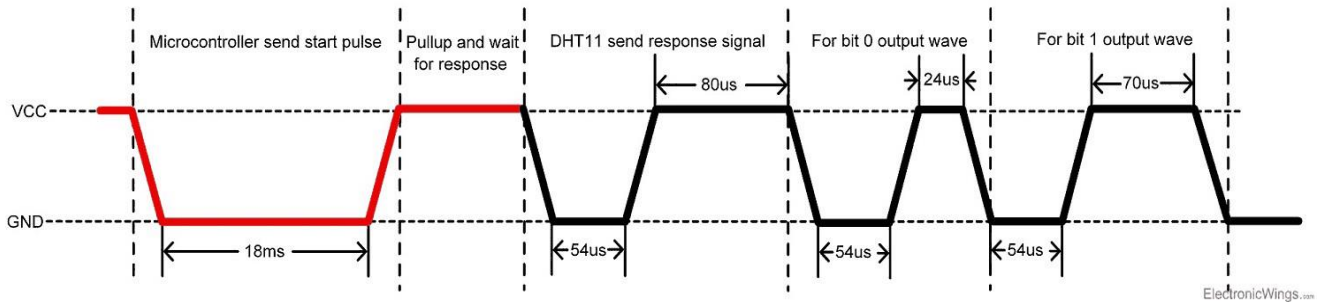
Pin Description:

Pin No.	Pin Name	Pin Description
1	VCC	Power supply 3.3 to 5.5 Volt DC
2	DATA	Digital output pin
3	NC	Not in use
4	GND	Ground

Communication with Microcontroller

DHT11 uses only one wire for communication. The voltage levels with certain time value defines the logic one or logic zero on this pin.

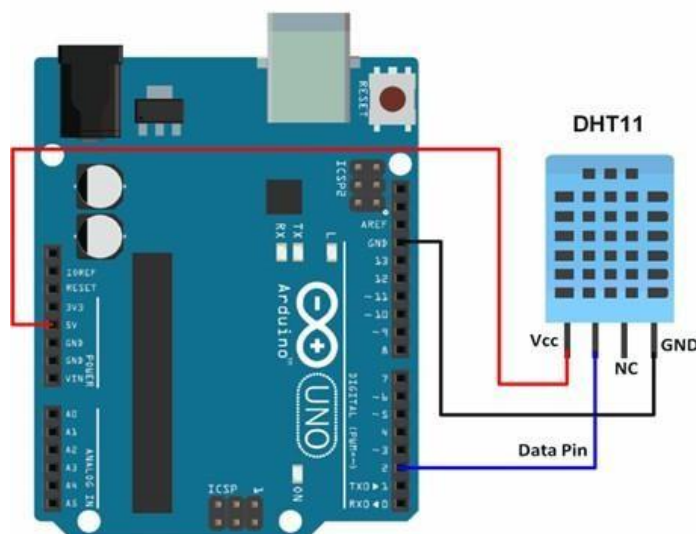
The communication process is divided in three steps, first is to send request to DHT11 sensor then sensor will send response pulse and then it starts sending data of total 40 bits to the microcontroller.



Specification of the DHT11 sensor

- Power supply: 3.3 to 5V DC
- Current consumption: max 2.5mA
- Operating range: 20-80% RH, 0-50°C
- Humidity measurement range: 20-90% RH
- Humidity measurement accuracy: $\pm 5\%$ RH
- Temperature measurement range: 0-50°C
- Temperature measurement accuracy: $\pm 2^\circ\text{C}$
- Response time: 1s
- Sampling rate: 1Hz (1 sample per second)
- Data output format: single-bus digital signal
- Data transmission distance: 20-30m (at open air)
- Dimensions: 15mm x 12mm x 5.5mm
- Weight: 2.5g
- Digital signal transmission protocol: 1 start signal + 40bit data + 1 checksum.

DHT11 Interfacing With Arduino



PERFORMANCE RESULT:

CONCLUSION:

Questions:

1. Which Arduino library is commonly used to interface with the DHT11 sensor?
2. Which Arduino pin is typically used to connect the DATA pin of the DHT11 sensor?
3. What function is used to initialize the DHT11 sensor in the Arduino code?
4. Write a step wise Procedure to connect DHT11 with Arduino Uno board
5. What does the function `dht.readTemperature ()` return?

[illegible]

SIGNATURE OF FACULTY WITH DATE

EXPERIMENT NO. 11

AIM: Demonstrate the working of Ultrasonic Distance Measurement sensor.

APPARATUS:

THEORY: The Ultrasonic sensor or HC-SR04 is used to measure the distance of the object using SONAR.

It emits the Ultrasound at a frequency of **40KHZ or 40000 Hz**. The frequency travels through the air and strikes the object on its path. The rays bounce back from the object and reach back to the module.

The four terminals of HC-SR04 are VCC, TRIG, ECHO, and GND. The voltage supply or VCC is +5V.

We can connect the ECHO and TRIG terminal to any of the digital I/O pin on the specific Arduino board.

The Ultrasonic sensors work best for medium ranges.

The resolution is 0.3cm.

The medium ranges of the sensor are 10cm to 3m. It works best at this duration.

The maximum range the sensor may detect is 4.5m.

Structure And Timing Diagram Of Ultrasonic Sensor:

We will set the TRIG pin to HIGH for some time (about 3 to 100 microseconds). As soon the TRIG pin is LOW, the Ultrasonic sensor sends the pulses and sets the ECHO pin to HIGH. When the sensor gets the reflected pulses, it sets the ECHO pin to LOW. We need to measure the time for which the ECHO pin was HIGH.

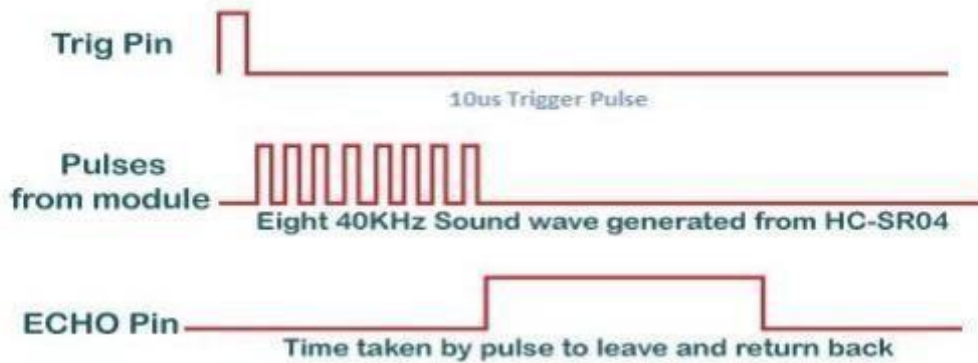
The timing diagram of the ultrasonic sensor HC-SR04 is shown below:



We will set the TRIG pin to HIGH for some time (about 3 to 100 microseconds). As soon the TRIG pin is LOW, the Ultrasonic sensor sends the pulses and sets the ECHO pin to HIGH. When the sensor gets the reflected pulses, it sets the ECHO pin to LOW. We need to measure the time for which the ECHO pin was HIGH

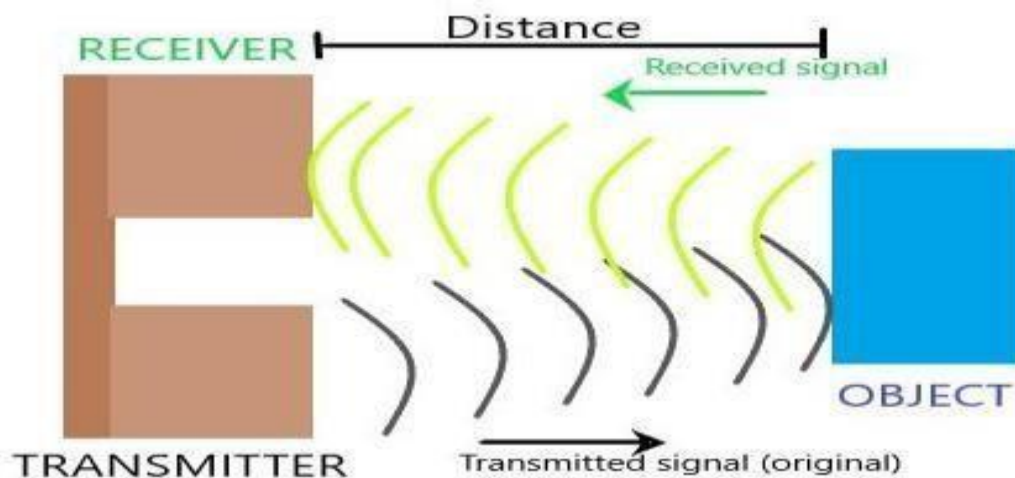
The timing diagram of the ultrasonic sensor HC-SR04 is shown below:

Ultrasonic HC-SR04 module Timing Diagram



We need to first set the TRIG (triggered) pin at HIGH. It will send out the burst of 8 cycles called the sonic burst, which will travel at the sound speed. It will be further received by the ECHO pin.

The time travelled by the sound wave is considered the ECHO pin's output time in microseconds.

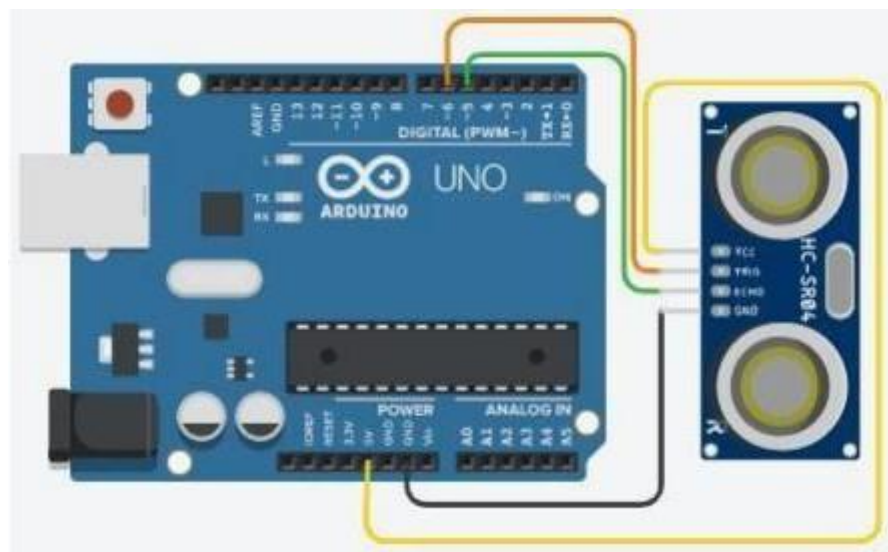


We will use the **PulseIn()** function to read the time from the output of the ECHO pin. It will wait for the specified pin to go HIGH and LOW.

The function would return the timing at the end. The TRIG pin is set LOW for 4 microseconds and then HIGH for 15 microseconds. The timing will be calculated in microseconds.

PROCEDURE:

- Open the Arduino IDE.
- Select the type of board from Tools -> Board -> Arduino UNO.
- Select the port from Tools -> Port -> COM.
- Connect the VCC pin of HC-SRO4 to 5V of the Arduino board.
- Connect the GND pin of HC-SRO4 to GND of the Arduino board.
- Connect the TRIG pin of HC-SRO4 to pin 9 of the Arduino board.
- Connect the ECHO pin of HC-SRO4 to pin 8 of the Arduino board.
- Upload the sketch to the sketch to the connection diagram.

CONNECTION DIAGRAM:**PERFORMANCE RESULT:**

CONCLUSION: _____

Q/A:

1. What is Ultrasonic Sensor and how does it works with Arduino?

2. Describe the pins of the Ultrasonic Sensor.

3. What is the maximum range of Ultrasonic sensor and can it detect multiple objects at once?

4. What exactly does the Ultrasonic sensor measures and how can we convert that parameter into the distance? _____

6. Can Ultrasonic sensor detect the height of the object or just the distance? ____

**SIGNATURE OF FACULTY WITH
DATE**

EXPERIMENT NO. 12

AIM: Verify the functionality of water flow sensor

APPARTUS:

THEORY: A water flow sensor is an electronic device that is used to measure water flow rate.

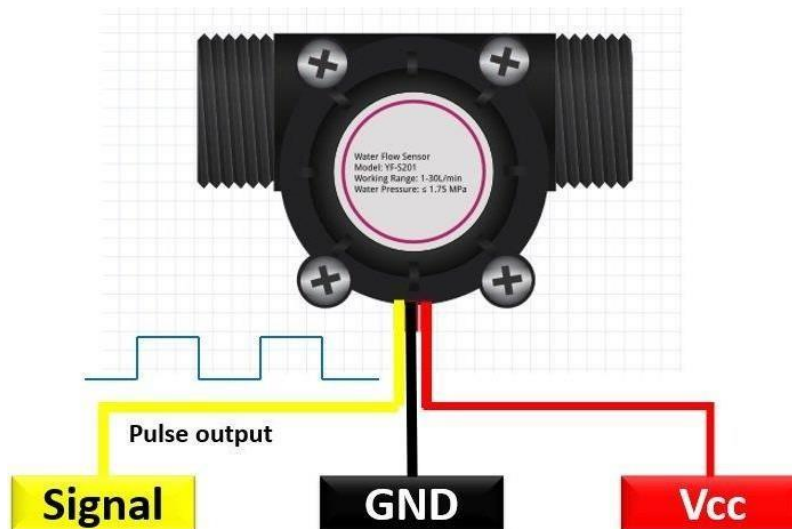
Furthermore, the flow rate means the volume of water passes through a sensor per unit of time. There are many types of water flow rate measurement sensors available in the market such as YF-B1, YF-B2, YF-B3, YF-B4, YF-B5, YF-B6, G1&2, G3&4, G1&8, and YF-S201.

All these sensors are almost the same except for the difference in flow rate range, operating voltage range, length, size, and material used. But their working principle and the procedure to interface with microcontrollers such as Arduino remains the same.

YF-S201 Water Flow Sensor Introduction

YF-S201 is a water sensor technically designed to measure the flow rate and volume of the desired fluid through the pipelines. It is a low-cost water flow sensor that consists of a copper body and water rotor.

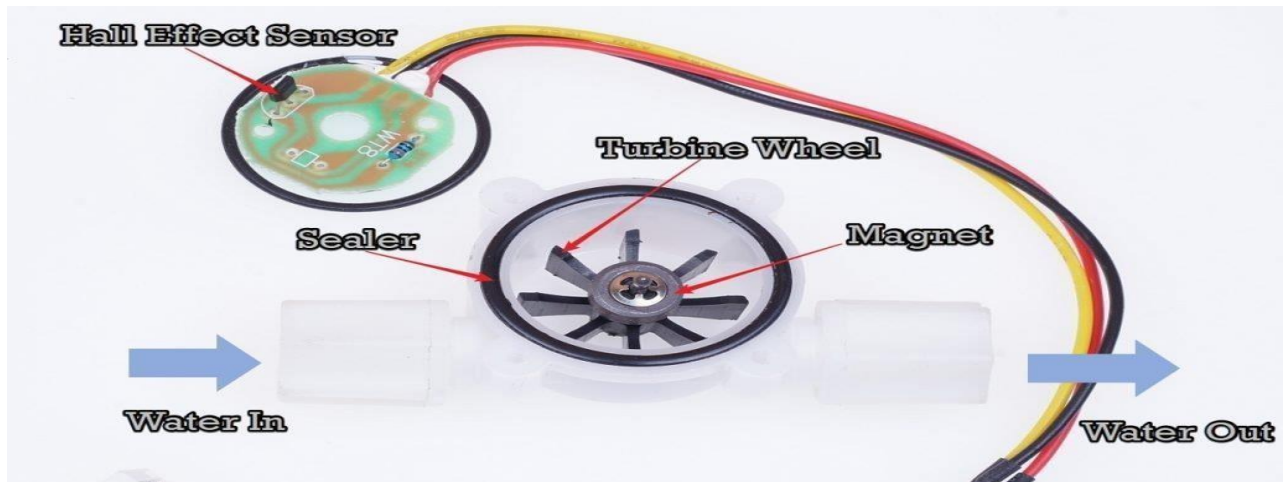
It also contains an internal circuit of the Hall effect sensor that works on the principle of electromagnetism and provides pulses at the output pin. It is a power-friendly and MCU compatible device with a flow rate of a maximum of 30 litres per minute. A small device with high accuracy finds its application from DIY projects to the industry for flow measurement.



- **How does Water Flow Sensor work?**

This water flow sensor consists of a water rotor (turbine wheel) and a Hall effect sensor. Water enters through one end and leaves through the other end of the sensor. When water flows through the flow sensor, it strikes the turbine wheel and the turbine wheel rotates. The speed of the turbine wheel has a direct relation with the speed of the flow of water through the water flow sensor. On each complete rotation of the turbine wheel, a hall effect sensor also produces a pulse that appears on the

signal output pin. In other words, the number of pulses that appear on the signal output pin is directly proportional to the rotational speed of the turbine.

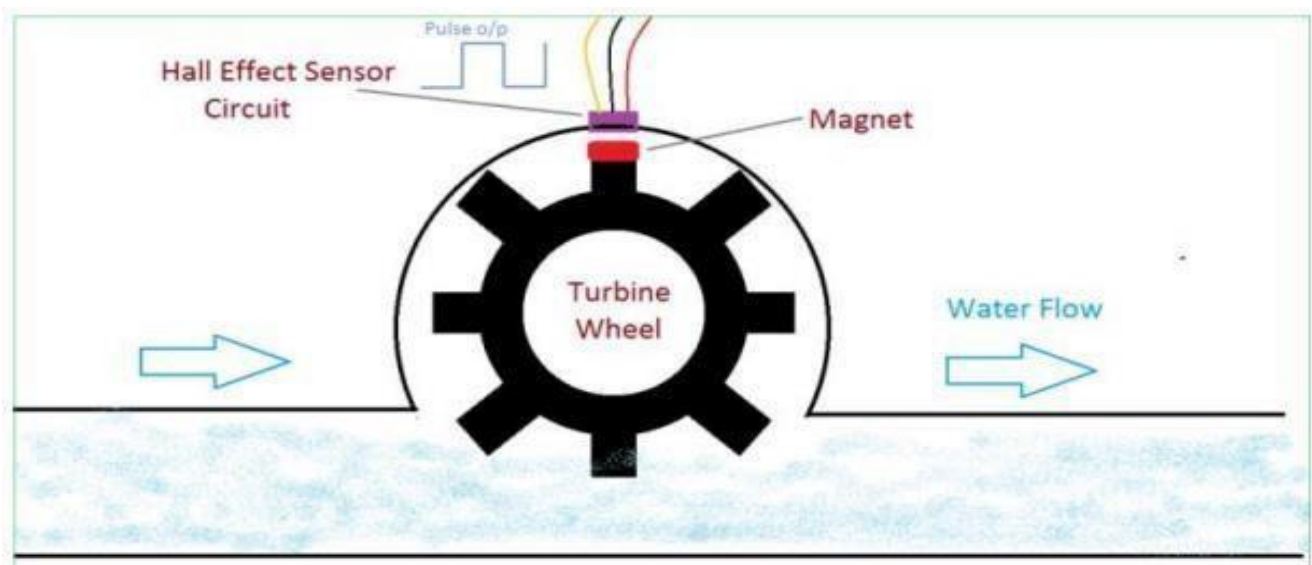


The YF-S201 is known as a Hall effect sensor because it operates on the Hall effect.

In order to measure the flow or volume of the fluid, we fix the sensor between the water inlet and outlet valves. When the fluid flows through, it rotates the rotor which has a magnet attached to it.

The speed interferes with the magnetic flux which is sensed by the Hall effect sensor and the sensor in return generates an output signal proportional to the magnetic flux with every revolution that the rotor makes.

As the water flow sensor is compatible with microcontrollers, we can observe the measurements on a computer serial monitor and can also display them on 16×2 LCD



In this flow meter, for every liter of liquid passing through it per minute, it outputs about 4.5 pulses. This is due to the changing magnetic field caused by the magnet attached to the rotor shaft.

We measure the number of pulses using an Arduino and then calculate the flow rate in litres per hour (L/hr) and total volume in Litre using a simple conversion formula.

- Pulse frequency = $7.5 \times \text{flow rate (ltr/min)}$
- Pulse frequency = $[7.5 \times \text{flow rate (ltr/hr)}]$

PROCEDURE:

- Open the Arduino IDE.
- Select the type of board from Tools -> Board -> Arduino UNO.
- Select the port from Tools -> Port -> COM.
- Connect the VCC pin YFS201 of to 5V of the Arduino board.
- Connect the GND pin of YFS201 to GND of the Arduino board.
- Connect Its Analog Pin To 2 Of Arduino.
- Upload the sketch to the connection diagram

PERFORMANCE RESULT

CONCLUSION:_____

—

Q/A:

1. What is YFS201 sensor and how does it work?

2. How do you connect YFS201 sensor with Arduino UNO board?

3. What are the key specifications of the sensor?

4. How does the YFS201 sensor's measurement range impact the maximum flow rate that it can detect, and how can thus be adjusted in the code?

5. What are some potential use of a YFS201 sensor and Arduino UNO in industrial automation or process control applications?

SIGNATURE OF FACULTY WITH DATE

EXPERIMENT NO. 13

AIM: Demonstrate the working of PIR Sensor using Arduino.

APPARATUS:

THEORY: PIR sensor is used for detecting infrared heat radiations. This makes them useful in applications involving detection of moving living objects that emit infrared heat radiations.

The output (in terms of voltage) of PIR sensor is high when it senses motion; whereas it is low when there is no motion (stationary object or no object).

How It Works?

The PIR sensor detects infrared light emitted by humans and animals. Commonly, it looks for changes in the detected infrared light, which means someone or something has moved within its detection field. When motion is detected, the sensor outputs a HIGH signal. This signal can be read by the Arduino, which in turn, activates the LED.



PIR Sensor

All living objects, whose body temperature is more than 0°C, emit the heat in form of infrared radiation through their body, also called as thermal radiations. This Radiated energy is invisible to human eye. These Signals can be detected by using PIR sensor which is specially designed for such purpose.



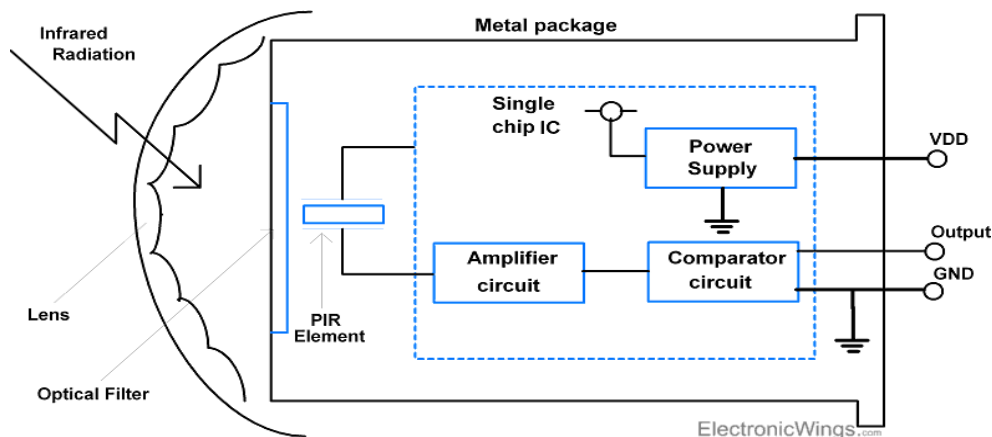
Grid eye illusion

- **PIR sensor i.e. Passive Infrared Sensor**, passive word indicates PIR Sensor does not generate or radiate any energy for detection purposes.
- PIR Sensors don't detect or measure "**HEAT**"; they detect the infrared radiation emitted or reflected from objects. They are small, inexpensive, low power and easy to use. They are commonly found at home, medical, factories etc. areas.

Specification of PIR Sensor

- Detection range: up to 7 meters
- Detection angle: 110 degrees
- Operating voltage: DC 4.5V - 12V DC
- Output signal: 3.3V digital output
- Delay time: adjustable from 0.3 seconds to 5 minutes
- Operating temperature: -15°C to +70°C
- Sensitivity: Adjustable

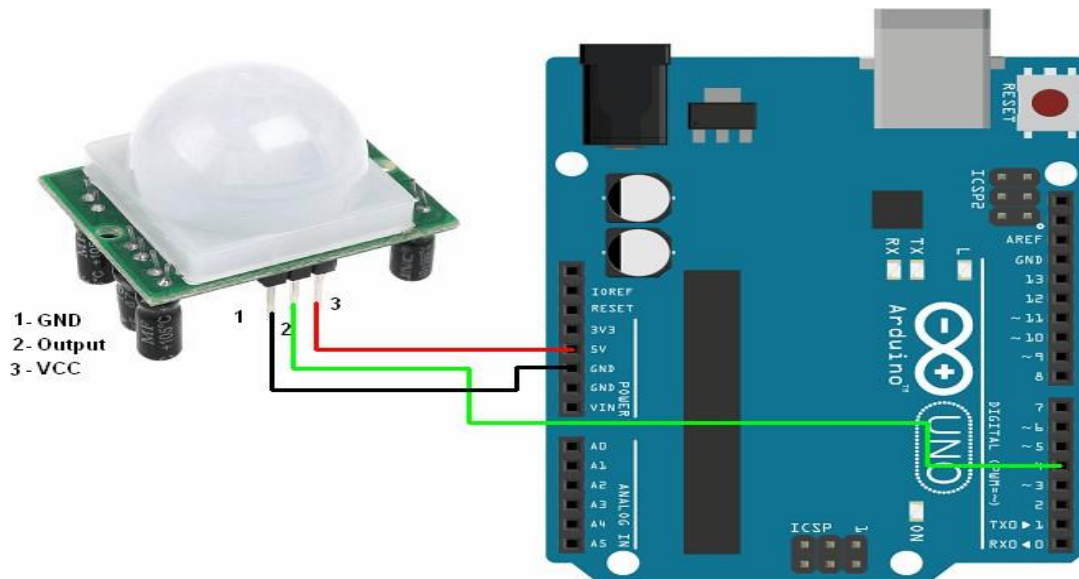
How PIR Sensor Work



PIR Sensor Blocks

- PIRs are basically made of a pyroelectric sensor, which can detect levels of infrared radiation.
- Above figure of PIR element shows the round metal can with a rectangular crystal in the center.
- Every object emits some low-level radiation, and the hotter objects emit more radiations.

PIR Sensor Connection with Arduino Board



PERFORMANCE RESULT:

EXPERIMENT NO. 14

AIM: Project based on electronics components and sensors

Title:

Apparatus:

Circuit diagram:

Working principle:

Performance Result:

Applications:

SIGNATURE OF FACULTY WITH DATE